

PART 1

The Context of Ecotourism and Environmental Sustainability

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Chapter 1

Ecotourism and Environmental Sustainability: An Introduction

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Ecotourism: Its Meaning and Significance

Thanks to the creation of previously inaccessible and undeveloped destinations, and a preference for independent and special interest holidays in non-resort locations, there is little left of the natural world that has not been exploited or commoditized for tourist consumption. The extension of the so-called ‘pleasure periphery’ (after Turner and Ash 1975; see Figure 1.1) into ever more remote and exotic areas has been driven to a significant degree by various forms of nature-based tourism, especially ecotourism, which are manifest in a range of environments from polar to tropical and terrestrial to aquatic, and which exhibit a strong correlation with peripheral and (predominantly public) protected areas. Nature-based tourism in general is one of the fastest growing sectors within the global tourism industry (Buckley 2000, Ryan et al. 2000, Wight 2001, Kuo 2002), it being defined as tourism ‘primarily concerned with the direct enjoyment of some relatively undisturbed phenomenon of nature’ (Valentine 1992: 108). Market estimates are hard to come by, given the lack of consensus over the use of the term(s), but it was suggested in 2004 that eco-/nature tourism was growing three times faster globally than the tourism industry as a whole (WTO 2004, cited in TIES 2006). Reasons for this growth include demographic changes in source countries (such as older populations and, in turn, the growing number of more experienced travellers), ‘beach boredom’ as a symptom of a maturing market for 3S (sun, sea, sand) holidays and increasing environmental awareness on the part of the general public (Ayala 1996). It is also attributable to the rapid development of an industry comprised of specialized (for example ecolodges, ecotour operators, and suppliers of transport services and infrastructures within a given ecotourism destination) and non-specialized businesses (such as hotel chains, airline and cruise ship operations, and retail travel agents), ranging from small- and medium-sized enterprises to transnational corporations (Weaver and Lawton 2007).

Under the broad banner of nature-based tourism, ecotourism has been suggested as a way in which increasing numbers of visitors seeking an intrinsically environmental tourism experience can be accommodated, whilst minimizing the costs and enhancing the benefits associated with natural area tourism (Boo 1990, Cater and Lowman 1994). Many definitions and types of ecotourism

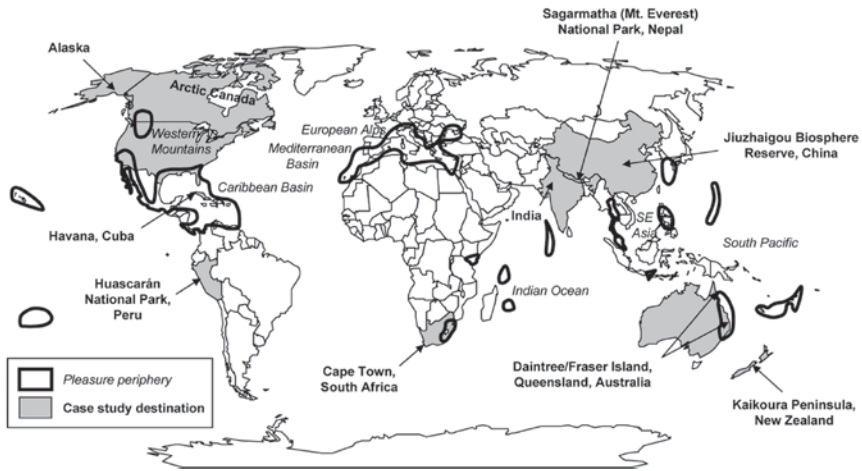


Figure 1.1 Location of the ‘pleasure periphery’, and of destinations and destination regions that feature as case studies in this book

Source: Adapted from Weaver (1998: 48)

exist (Orams 2001, Donohoe and Needham 2006) but, in its broadest sense, it concerns travel to a natural area; involving local people; feeding economic profit into local environmental protection; and contributing to the maintenance of the local environment and species diversity through minimizing visitor impact and promoting tourist education (Valentine 1993, Western 1993, Ceballos-Lascuráin 1998, Diamantis 1999, Fennell 2001, 2003). In this way, ecotourism (as a subset of alternative tourism) is being promoted by governments and the tourism industry as a sustainable alternative to mass tourism. However, critics have suggested that ecotourism can be damaging to the natural environment, not least with respect to the environmental cost of air travel to popular destinations and the potential for unwittingly disturbing soils and ecosystems through accommodation and activities on site (Wheeller 1991, 1993, Hjalager 1996, Conservation International 1999, Wearing and Neil 1999, Kruger 2005). Doubts have also been raised about the ‘true’ motivations of ecotourists, which appear to have as much to do with sustaining the ego as the environment (Wheeller 1993), and the sense in which ecotourism encourages increased use of natural areas and greater penetration into sensitive environments, thereby putting the very future of indigenous tourism industries at risk (Mihalic 2000). There is a tendency to overstate its significance, too, when in fact it constitutes only a small proportion of global tourism (estimated at between 2 per cent and 4 per cent; WTO 2002, cited in Cater 2004).

Before we explain the aim, philosophy and structure of this book, it is perhaps useful to provide a brief introduction to definitions of ecotourism; ecotourist types and motivations; and ecotourism’s (environmental) sustainability credentials.

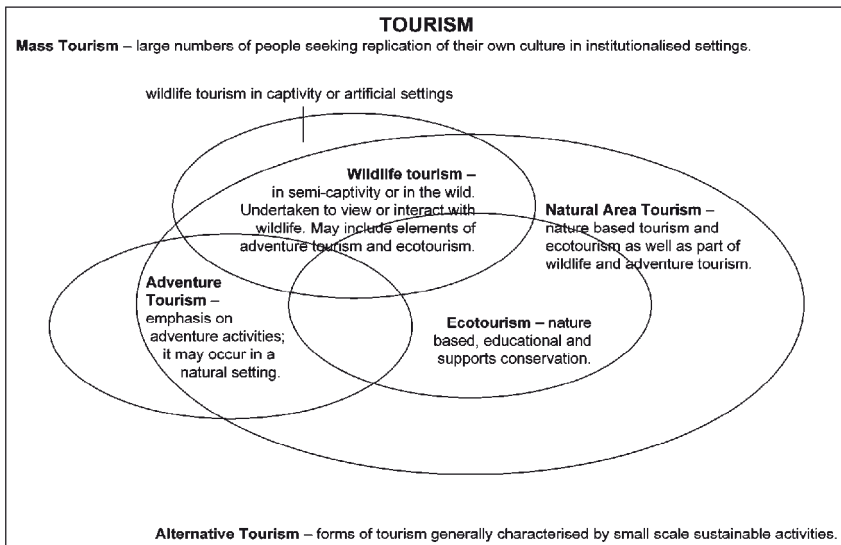


Figure 1.2 Relationship of ecotourism to other forms of tourism

With regards to the first consideration, Fennell's (2001) content analysis of ecotourism definitions identifies no less than 85 of these in the literature available at that time which, for a term that only entered academic discourse in the late 1980s, is quite remarkable (if not a little counter-productive). Definitions of ecotourism leave much to the interpretation of the reader, but they more or less cohere around three criteria, namely that '(1) attractions should be predominantly nature-based, (2) visitor interactions with those attractions should be focused on learning or education, and (3) experience and product management should follow principles and practices associated with ecological, socio-cultural and economic sustainability' (Weaver and Lawton 2007: 170). However, given the existence of something that approximates to an ecotourism experience in modified settings such as botanic gardens (see Chapter 12), not to mention the 'accidental ecotourist' and the potential for behaviour associated with states of mindlessness as well as mindfulness (after Moscardo 1996, 1999), plus an emerging critique of earlier work which problematizes ecotourism's role in sustainable development and environmental conservation (for example Butcher 2005, Kruger 2005, Cusack and Dixon 2006, Southgate 2006), one has to accept that there is no definition available that satisfactorily explains this phenomenon of contemporary (postmodern) society. That said, the distinction may usefully be made here between positive (i.e. what is) and normative (i.e. what should be) ecotourism definitions; whereas a universal definition that everyone can subscribe to remains a distant prospect (and, arguably, a futile project), there is an emerging consensus over the inclusion of certain 'value-based dimensions' such as conservation, community involvement

and social responsibility (Weaver and Lawton, 2007: 1169). This is also apparent in Garrod's (2003) Delphi study, which indicates a preference amongst experts for prescriptive, rather than descriptive, definitions of ecotourism (as well as summarizing the reasons why they can be useful).

Figure 1.2 illustrates the relationship between ecotourism and other forms of tourism. This follows the typology of Newsome et al. (2002) in which ecotourism is seen entirely as a subset of natural area tourism, incorporating elements of wildlife and adventure tourism as acknowledged by a number of researchers (for example Page and Dowling 2002, Fennell 2003). It accommodates Fennell's (2003: 16) belief 'that ecotourism is distinct from mass tourism and various other forms of AT [alternative tourism]', whilst acknowledging the common ground between them (see Weaver 2005 on mass ecotourism, and Fennell 2003 on the difficulty of distinguishing between adventure tourism, cultural tourism and ecotourism in similar environmental settings). However, by its inclusion we are not asserting that this is the only way of seeing ecotourism in the context of other tourism types (see Weaver 2001 for alternative illustrations).

Moving on to the second of the above-mentioned considerations, namely ecotourists and their motivations, empirical research has shown that those who have participated in some form of nature-based tourism (which qualifies them as 'ecotourists' in a number of published studies) tend to be slightly older (between 35 and 54 in particular), better educated and more affluent (and, therefore, prepared to pay more for their holidays) than those who do not participate (see Wilson and Garrod 2003 for a summary of demographic and other characteristics). Furthermore, they are likely to stay longer and be more tolerant of basic conditions than general travellers. Environmental awareness and respect for local customs and culture are also differentiating features, as is gender (at least for certain activities such as bird watching, which attracts more males than females). However, despite having much in common, there are many different types of eco-/nature tourist that are distinguishable by factors including level of organisation (such as do-it-yourself ecotourists, ecotourists on tours and school or scientific groups, as identified by Kusler 1991) and dedication to nature (for example Lindberg's 1991 typology, comprising hard-core, dedicated, mainstream and casual nature tourists). Similarly, the degree of difficulty and level of engagement are used by Laarman and Durst (1987) in arguing for the distinction between 'hard' and 'soft' ecotourism experiences.

The most common reasons offered by ecotourists for undertaking a given trip are a desire to enjoy scenery and nature, and to encounter new places and experiences (see Chapters 9 and 11), with the imperative to 'get away from it all' of less importance to the ecotourist than, say, the adventure tourist. Motivations can and do vary by market, however, as summarized by Wight (2001). Suffice to say, for an *explanation* as to what motivates the ecotourist, one sometimes has to look beyond the academic literature. For example, in the first quarter of 2003 the British Broadcasting Corporation (BBC) conducted a survey of visitors to its website that attempted to establish the '50 things to do before you die', the

Table 1.1 ‘50 things to do before you die’

1	<i>Swim with dolphins</i>	11	Walk the Inca Trail	21	Travel on the Orient Express	31	Gallop a horse along a beach	41	Fish for Blue Marlin in the Florida Keys
2	<i>Dive the Great Barrier Reef</i>	12	Climb Sydney Harbour Bridge	22	<i>Bathe an elephant</i>	32	Explore the pyramids by camel	42	Go paragliding
3	Fly on Concorde	13	Escape to a paradise island	23	<i>Visit Antarctica</i>	33	Travel on the Trans-Siberian Railway	43	Play a round of golf at Augusta, Georgia
4	<i>Go whale-watching</i>	14	Drive a Formula 1 Car	24	Ride a Harley Davidson on the open road	34	Watch the sunset at Uluru (Ayers Rock)	44	<i>Trek to see a mountain gorilla</i>
5	<i>Dive with sharks</i>	15	Go white-water rafting	25	Have a go at cowboy ranching	35	Go wing-walking	45	<i>See tigers in the wild</i>
6	Skydiving	16	Walk the Great Wall of China	26	Climb Mount Everest	36	Climb Mount Kilimanjaro	46	Hurtle down the Cresta Run
7	Fly in a hot air balloon	17	Bungee jumping	27	Experience a waterfall	37	Fly over a volcano	47	Visit Walt Disney World, Florida
8	Fly in a fighter jet	18	Take a train journey across the Canadian Rockies	28	See the Earth from space	38	Drive a husky sled	48	Visit Las Vegas casinos
9	<i>Go on safari</i>	19	Drive along Route 66	29	<i>Explore the Galapagos Islands</i>	39	Hike up a glacier	49	<i>Watch Orang-Utans</i>
10	See the Northern Lights	20	Fly in a helicopter over the Grand Canyon	30	<i>Trek through a rainforest</i>	40	Ride a rollercoaster	50	<i>Go polar bear watching</i>

Source: BBC (2003)

Note: Experiences with a significant ecotourism component are highlighted in *italics*.

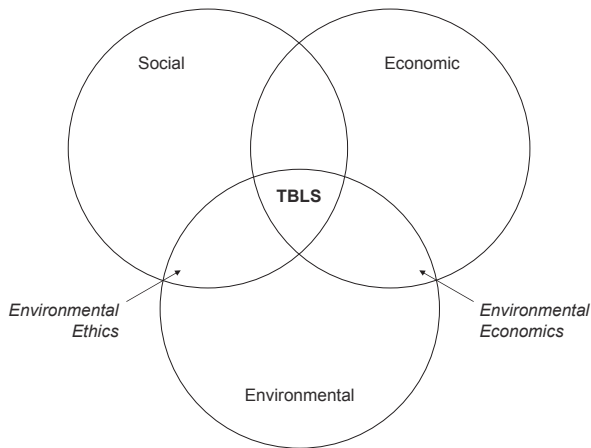


Figure 1.3 Elements of triple-bottom-line-sustainability (TBLS)

results of which were subsequently broadcast on the BBC's *Holiday* programme (Table 1.1). Not surprisingly, long-distance travel features in nearly all of these 'must-do' activities. What is more pertinent, however, is that a large number correlate with a wildlife or adventure tourism experience, most likely with a significant ecotourism component. Examples include, but are not confined to: swimming with dolphins (1); going on safari (9); seeing elephants in the wild (22); exploring the Galapagos Islands (29); trekking through a rainforest (30); and watching mountain gorillas (44). Crucially, these activities remind us of the potential for ecotourism (and related forms of tourism) to deliver what Maslow, writing in the 1950s, described as a 'peak experience'. In turn, they are able to address higher order human needs, especially self-actualization and transcendence (personal growth and fulfilment, and helping others to self-actualize). Indeed, opportunities to develop knowledge and understanding in beautiful surroundings (and, therefore, to satisfy cognitive and aesthetic needs as a pre-condition for self-actualization) are present in most, if not all, ecotourism encounters. Hence, we can conclude that ecotourists are motivated by so-called 'growth motivators' (and not just 'deficiency motivators', such as physiological, security, belongingness/love and esteem needs), and that ecotourism can help them to achieve their full potential (see Maslow 1987 for an explanation of these terms).

Finally, there is the issue of ecotourism and environmental sustainability (the latter being one of the components comprising the 'triple-bottom-line') (see Figure 1.3). Broadly speaking, ecotourism is linked to the emergence of 'new tourism' and associated production practices and consumption patterns (see Poon 1993); it is one of four key types identified by Shaw and Williams (2004: 119), the others being heritage/cultural tourism, adventure tourism, and visiting theme parks/mega-shopping malls (see also Mowforth and Munt 2003: 93 for an extended

typology/glossary of terms). Although contested, it is suggested that some of these new forms of tourism are based around sustainable ideas (such as environmental stewardship, inter-generational and intra-generational equity). Certainly, we are told that ecotourism is small in scale, non-consumptive, ethical/responsible, and of benefit to local people. It is also thought to encourage pro-environmental behaviours (both home and away), when accompanied by interpretation (see Chapter 11).

This was the thinking behind the International Year of Ecotourism, in 2002 (as proclaimed by the United Nations). However, in the absence of an appropriate management regime, where ‘those who initiate and develop ecotourism ... operate within the capacity of the environment to absorb the impacts’ (Wilson and Garrod 2003: 4), ecotourism is unlikely to be sustainable ecologically by any relevant measure. There is also the possibility that ecotourists (however well intentioned) might disturb the feeding and breeding patterns of wildlife, transmit diseases and modify habitats, just by being present in environmentally sensitive areas. In addition, a great many tourism products labelled with the prefix ‘eco-’ have few, if any, of the above characteristics (see Chapter 13), which suggests that ‘ecotourism is being used [primarily] to meet economic objectives by promoting the quality of the environment to attract international tourists’ (Holden 2005: 130), and not for the specific purpose of natural resource conservation (a minority of cases excepted). Finally, given that most ecotourism destinations are geographically remote in relation to the markets they serve, there is the issue of carbon expenditures associated with long-distance travel and their contribution to climate change (see Chapters 2 and 3). Clearly, then, ecotourism is not (and can never be) *synonymous* with environmental sustainability, contrary to what some advocates would have us believe.

The Ecotourism Body of Knowledge

Despite the title of this section, it is not our intention here to review comprehensively the literature on ecotourism, especially as some excellent reviews exist already (see Weaver and Lawton 2007, referenced throughout this opening chapter). Instead, we direct readers to the following:

- The ever-expanding range of introductory textbooks, which deal at length with the origins and concepts of ecotourism; where it lies within broader tourism types; its economic, environmental/ecological and socio-cultural impacts; the planning and management of ecotourism businesses; and ecotourism organizations and policies (see, for example, Wearing and Neil 1999, Page and Dowling 2002, Fennell 2003, Weaver 2008).
- More specialized publications that focus on certain aspects such as policy and planning (Fennell and Dowling 2003); underlying environmental ideologies linked to politics of north–south relations and social justice

(Duffy 2002, Butcher 2007); and economic and sustainable development in less developed countries (Weaver 1998, Honey 1999).

- The 500-plus articles of relevance in refereed journals, accessible via academic databases and search engines using appropriate keywords such as 'ecotourism', 'ecotourist' and so on (including those published in the dedicated *Journal of Ecotourism*, established in 2002).

In their review of the ecotourism literature, Weaver and Lawton (2007) identify three 'macro-themes' that may help contextualize the contents and approach of this book. Firstly, they note that the subject has been transformed over the past decade due to processes of de-differentiation (the blurring of the boundaries between ecotourism and other forms of tourism, as exemplified by upscale ecolodges and recreational hunting and fishing in protected areas) and segmentation (such as the emergence of whale watching and Antarctic tourism as sub-fields of enquiry). Secondly, they record ongoing efforts to make sense of ecotourism impacts, which follow three distinct research trajectories (four, in their original schema) addressing ecological, economic and ethical considerations (dubbed 'the 3Es' by Howie 2003). When arranged in this order, these form a continuum that at the one end is dominated by the natural sciences and their preferred methods of investigation, and at the other end the social sciences. Finally, they identify a profound 'North-South' (or, alternatively, 'West-Rest') division in both the sector and the literature. This is confirmed by the tendency for supply-side case studies (of venues, businesses, etc.) to be based in the developing world, in particular Central America, Sub-Saharan Africa and Southeast Asia (see Weaver 1998 for details), and for demand-side case studies (of markets, products, etc.) to be based in the developed world, from which much knowledge of ecotourism originates (the implications of which are explored in Cater 2006).

Organisation of the Book

With the contextual discussion complete, we now turn to the remainder of this book, which aims to offer conceptual and practical insights into the complex interactions between ecotourism and heterogeneous and dynamic natural environments (building on the foundation provided by the sum of the contributions to Buckley 2004). Although privileging environmental relationships (encompassing physical environment, flora and fauna), consideration is given to government policy, marketing by suppliers, consumer behaviour and visitor/environmental management as factors that mediate tourist interactions with the natural environment. Some of these factors correlate with 'strategically important' areas of ecotourism that have been neglected thus far in the literature (see Weaver and Lawton 2007).

The book is divided into three parts. The first part, comprising Chapters 1 to 3, introduces ecotourism and tourism-environment relationships at the local

and global scales. The second part, comprising Chapters 4 to 12, is devoted to a diverse series of case studies from around the world (see Figure 1.1), incorporating analysis from a range of geographic scales and integrating natural– and social–scientific knowledge, yet resisting a noticeable trend to showcase best practice (given that this is rarely transferable from one context to another). A variety of ecotourism activities, ecosystem types, ecosystem components and environmental responses are examined in these chapters. The final part, comprising Chapters 13 and 14, draws together some of the lessons revealed by earlier chapters in a critical examination of eco-/nature tourism and its articulation with discourses of environmental sustainability, and includes an ‘agenda’ for future research in this area.

In Chapter 2, Andrew Holden considers the negative and positive impacts of tourism in general (not just ecotourism) on the natural environment, emphasizing the significance of human agency and of the values we attach to nature (which vary by cultural group and over time) in determining the net environmental impact of tourism. The roles of environmental economics and environmental ethics in promoting natural resource conservation are also discussed in this chapter. It provides a basis for subsequent chapters that explore the relationship of tourism with natural resources in theory and offer practical examples of resource conserving behaviour vis-à-vis resource destructive behaviour in eco- and related forms of tourism. Then in Chapter 3, Colin Hunter uses Ecological Footprint (EF) analysis, which has the ability to compare impacts generated in transit with those at the destination, to assess the environmental sustainability of ecotourism relative to mass tourism. He makes the critical distinction between gross and net EF estimates (with the latter taking into account the footprint that would have been generated had the tourist stayed at home), in outlining the circumstances in which return trips from some of the world’s leading source countries/cities to recognized ecotourism destinations could be said to compare favourably with equivalent trips to mass market destinations (despite concerns about ecotourism’s frequent reliance upon aviation).

The next nine chapters provide case studies of particular sites and destinations. In Chapter 4, Alton Byers undertakes a comparative analysis of landscape change in the alpine zones of Sagarmatha (Mt. Everest) National Park, Khumbu, Nepal and the Huascarán National Park, Cordillera Blanca, Peru. He attributes the degradation of the alpine zone within each national park to poorly controlled adventure tourism and associated impacts, and reports on progress made with regards to conserving, protecting and restoring these fragile environments. Tourism in protected areas also features in Chapter 5, by Wenjun Li. Using a case study of Jiuzhaigou Biosphere Reserve, located within China’s Sichuan Province, she underlines the importance of community involvement in tourism and in ensuring that residents benefit financially from it, here by means of a ‘joint-stock mechanism’. With such a mechanism in place, the likelihood of balancing resource conservation and economic development objectives through tourism is increased. Then in Chapter 6, Emma Stewart and Dianne Draper investigate the development of cruise tourism

in Arctic Canada and the environmental consequences thereof. They identify a package of measures aimed at ensuring a successful and sustainable future for the Canadian Arctic as a cruise tourism destination, from the perspective of the stakeholders involved (communities, operators and tourists), and they discuss the likely implications of climate change for the industry in this part of the world.

The focus switches to wildlife tourism and particular species in the four chapters that follow. In Chapter 7, Terry DeBruyn and Tom Smith explore some of the issues raised by the growing demand for brown bear-viewing on Alaska's public lands, together with resource management interventions designed to minimize human impacts on bears and associated wilderness. This culminates in a series of guidelines for managing bear-viewing areas that prioritizes bear conservation, human safety and visitor satisfaction. Then, in Chapter 8, Simon Evans critically examines the role of ecotourism in protecting the wild tiger populations of India. He voices concern over the widespread practice of encouraging tigers to visit a site habitually through baiting, with a view to guaranteeing sightings. This situation is compounded by marketing approaches that heighten tourists' expectations of viewing and photographing tigers at close range (see Dobson's 2007 study of sharks for some interesting parallels) and, more seriously, declining tiger numbers as a consequence of large-scale poaching (which reduces the likelihood of a chance encounter). In contrast to the experience outlined in Chapter 5 (see above), he notes a tendency for local populations to be excluded from ecotourism and conservation activities, leading to the formation of eco-enclaves (the tiger reserves and the hotels located outside reserve boundaries) which capture most of the financial benefits of tiger-related tourism. Following this, in Chapter 9 Leah Burns draws on the testimony of visitors and park rangers, in a study of tourism impacts and tourist aspirations in relation to the dingoes of Fraser Island, Queensland, Australia. She evaluates hard (e.g. fences) and soft (e.g. visitor education) approaches to managing human-dingo interactions, in the wake of highly publicized incidents that resulted in people being harmed and dingoes being culled (and, like Simon Evans before her, draws attention to the problem of habituation in wildlife tourism).

The role and potential of interpretation provision in managing visitor behaviour and attitudes towards wildlife and ecosystem conservation is examined in the next two chapters. In Chapter 10, Davina Stanford uses a large-sample survey of visitors ($n=372$) to the fur seal colony at Kaikoura Peninsula, New Zealand to evaluate the effectiveness of six different messages designed to encourage the visitor not to get too close to the seals (less than ten metres), which correspond to Kohlberg's stages of moral development. The results indicate that stage four and five messages (emphasizing good citizenship, and the impacts and consequences of less than desirable behaviour) are the most influential. Then in Chapter 11, Jennifer Hill and Georgie Gough report the findings of a visitor survey conducted at Crocodylus Rainforest Village in Daintree, Australia ($n=267$). This provides insights into interpretation provision as a tool for educating the travelling public about biodiversity, including pertinent threats and conservation issues, and suggests that it engenders (or at the very least consolidates) positive attitudes and

behavioural intentions towards ecosystem conservation specifically and to the environment more generally, as well as enhancing visitor satisfaction.

If the last two chapters (excluding the conclusion) have something in common, it is that they deconstruct the term 'ecotourism', firstly by extending it to encompass modified spaces and, secondly, by dismissing it as a mere marketing label (so-called 'greenwashing'). In Chapter 12, Lucy A. Sutherland examines the potential role of botanic gardens in less developed countries (here represented by South Africa and Cuba) in linking into important urban green space and wildlife corridor networks, and in alleviating pressure on adjacent protected areas by providing a substitute ecotourism experience. Using a combination of questionnaires distributed to and semi-structured interviews with botanic garden visitors and inbound tour operators, she identifies the factors that, in combination, will determine whether or not these attractions can be incorporated within ecotourism programmes (as a more sustainable alternative to visiting fragile natural environments). This is followed by Ken Simpson's polemical essay, in Chapter 13, on the theoretical foundations of ecotourism and the political, societal and ecological realities that militate against them. He challenges the notion that 'if it's ecotourism, it must be sustainable' by arguing that the cost-benefits associated with any given form of tourism are the product of complex interactions between a destination with experiences to sell and a multi-faceted market with the resources to buy, rather than being fixed or given. Indeed, it would seem that 'ecotourism' is in danger of becoming little more than a brand, due to its appropriation by tourism operators and destination marketers. To conclude, the key issues raised by the book are summarized in Chapter 14, and a number of directions for ecotourism (and environmental sustainability) research are proposed, informed by wider developments in tourism studies and the social sciences.

References

- Ayala, H. (1996) Resort ecotourism: a paradigm for the 21st century. *Cornell Hotel and Restaurant Administration Quarterly*, 37, 46–53.
- British Broadcasting Corporation (BBC) (2003) *50 Things To Do Before You Die*. Available at: <http://www.bbc.co.uk/50/destinations/things> [accessed: 19 September 2003].
- Boo, E. (1990) *Ecotourism: The Potentials and the Pitfalls* (Vol. 1). World Wildlife Fund, Washington, DC.
- Buckley, R. (2000) Tourism in the most fragile environments. *Tourism Recreation Research*, 25, 31–40.
- Buckley, R. (2003) *Case Studies in Ecotourism*. CABI Publishing, Wallingford.
- Buckley, R. (ed.) (2004) *Environmental Impacts of Ecotourism*. CABI Publishing, Wallingford.
- Butcher, J. (2005) The moral authority of ecotourism: a critique. *Current Issues in Tourism*, 8, 114–124.

- Butcher, J. (2007) *Ecotourism, NGOs and Development: A Critical Analysis*. Routledge, Abingdon.
- Cater, E. (2004) Ecotourism: theory and practice. In A.A. Lew, C.M. Hall and A.M. Williams (eds) *A Companion to Tourism*. Blackwell Publishing, Oxford, pp. 484–497.
- Cater, E. (2006) Ecotourism as a Western construct. *Journal of Ecotourism*, 5, 23–39.
- Cater, E. and Lowman, G. (eds) (1994) *Ecotourism: A Sustainable Option?* John Wiley & Sons, Chichester.
- Ceballos-Lascuráin, H. (1998) Introduction. In K. Lindberg, M. Epler-Wood and D. Engeldrum (eds) *Ecotourism – A Guide for Planners and Managers* (Vol. 2). The Ecotourism Society, North Bennington, Vermont, pp. 7–10.
- Conservation International (1999) *The Green Host Effect – An Integrated Approach to Sustainable Tourism and Resort Development*. Conservation International, Washington, DC.
- Cusack, D. and Dixon, L. (2006) Community-based ecotourism and sustainability: cases in Bocas del Toro Province, Panama and Talamanca, Costa Rica. *Journal of Sustainable Forestry*, 22, 157–182.
- Diamantis, D. (1999) The concept of ecotourism: evolution and trends. *Current Issues in Tourism*, 2, 93–122.
- Dobson, J. (2007) Shark! A new frontier in tourist demand for marine wildlife. In J. Higham and M. Lück (eds) *Marine Wildlife and Tourism Management: Insights from the Natural and Social Sciences*. CABI Publishing, Wallingford, pp. 49–65.
- Donohoe H.M. and Needham, R.D. (2006) Ecotourism: the evolving contemporary definition. *Journal of Ecotourism*, 5, 192–210.
- Duffy, R. (2002) *A Trip Too Far: Ecotourism, Politics and Exploitation*. Earthscan, London.
- Fennell, D. (2001) A content analysis of ecotourism definitions. *Current Issues in Tourism*, 4, 403–421.
- Fennell, D. (2003) *Ecotourism: An Introduction* (2nd edition). Routledge, London.
- Fennell, D. and Dowling, R.K. (eds) (2003) *Ecotourism Policy and Planning*. CABI Publishing, Wallingford.
- Garrod, B. (2003) Defining marine ecotourism: a Delphi study. In B. Garrod and J.C. Wilson (eds) *Marine Ecotourism: Issues and Experiences*. Channel View Publications, Clevedon, pp. 17–36.
- Hjalager, A. (1996) Tourism and the environment: the innovation connection. *Journal of Sustainable Tourism*, 4, 201–218.
- Holden, A. (2005) *Tourism Studies and the Social Sciences*. Routledge, London.
- Honey, M. (1999) *Ecotourism and Sustainable Development: Who owns Paradise?* Island Press, Washington, DC.
- Howie, F. (2003) *Managing the Tourist Destination*. Continuum, London.

- Kruger, O. (2005) The role of ecotourism in conservation: panacea or Pandora's box? *Biodiversity and Conservation*, 14, 579–600.
- Kuo, I-L. (2002) The effectiveness of environmental interpretation at resource-sensitive tourism destinations. *International Journal of Tourism Research*, 4, 87–101.
- Kusler, J.A. (1991) Ecotourism and resource conservation: introduction to issues. In J.A. Kusler (ed.) *Ecotourism and Resource Conservation: A Collection of Papers* (Vol. 1). Omnipress, Madison, WI.
- Laarman, J. and Durst, P. (1987) *Nature Travel and Tropical Forests. FREI Working Paper Series*. Southeastern Center for Forest Economics Research, North Carolina State University, Raleigh, NC.
- Lindberg, K. (1991) *Policies for Maximising Nature Tourism's Ecological and Economic Benefits*, World Resources Institute, Washington, DC.
- Maslow, A.H. (1987) *Motivation and Personality* (3rd edition). Harper & Row, New York, NY.
- Mihalic, T. (2000) Environmental management of a tourist destination: a factor of tourism competitiveness. *Tourism Management*, 21, 65–78.
- Moscardo, G. (1996) Mindful visitors: heritage and tourism. *Annals of Tourism Research*, 23, 376–397.
- Moscardo, G. (1999) *Making Visitors Mindful*. Sagamore Publishing, Champaign, IL.
- Mowforth, M. and Munt, I. (2003) *Tourism and Sustainability: New Tourism in the Third World* (2nd edition). Routledge, London.
- Newsome, D., Moore, S.A. and Dowling, R.K. (2002) *Natural Area Tourism: Ecology, Impacts and Management*. Channel View Publications, Clevedon.
- Newsome, D., Dowling, R.K. and Moore, S. (2005) *Wildlife Tourism*. Channel View Publications, Clevedon.
- Orams, M.B. (2001) Types of Ecotourism. In D.B. Weaver (ed.) *The Encyclopaedia of Ecotourism*. CABI Publishing, Wallingford, pp. 23–36.
- Page, S.J. and Dowling, R.K. (2002) *Ecotourism*. Prentice Hall, Harlow.
- Poon, A. (1993) *Tourism, Technology and Competitive Strategies*. CABI Publishing, Wallingford.
- Ryan, C., Hughes, K. and Chirgwin, S. (2000) The gaze, spectacle and ecotourism. *Annals of Tourism Research*, 27, 148–163.
- Southgate, C.R.J. (2006) Ecotourism in Kenya: the vulnerability of communities. *Journal of Ecotourism*, 5, 80–96.
- Shaw, G. and Williams, A.M. (2004) *Tourism and Tourism Spaces*. Sage, London.
- The International Ecotourism Society (TIES) (2006) *Fact Sheet: Global Ecotourism*. Available at: <http://www.box.net/shared/static/eaukonjc0a.pdf> [accessed: 13 August 2008].
- Turner, L. and Ash, J. (1975) *The Golden Hordes: International Tourism and the Pleasure Periphery*. Constable, London.

- Valentine, P.S. (1992) Review. Nature-based tourism. In B. Weiler and C.M. Hall (eds) *Special Interest Tourism*. Belhaven Press, London, pp. 105–128.
- Valentine, P.S. (1993) Ecotourism and nature conservation: a definition with some recent developments in Micronesia. *Tourism Management*, 14, 107–115.
- Wearing, S. and Neil, J. (1999) *Ecotourism: Impacts, Potential and Possibilities*. Butterworth Heinemann, Oxford and Melbourne.
- Weaver, D.B. (1998) *Ecotourism in the Less Developed World*. CABI Publishing, Wallingford.
- Weaver, D.B. (2001) Ecotourism in the context of other tourism types. In D.B. Weaver (ed.) *The Encyclopaedia of Ecotourism*. CABI Publishing, Wallingford, pp. 73–83.
- Weaver, D.B. (2005) Mass and urban ecotourism: new manifestations of an old concept. *Tourism Recreation Research*, 30, 19–26.
- Weaver, D. (2008) *Ecotourism* (2nd edition). John Wiley & Sons, Chichester.
- Weaver, D.B. and Lawton, L.J. (2007) Twenty years on: the state of contemporary ecotourism research. *Tourism Management*, 28, 1168–1179.
- Western, D. (1993) Defining ecotourism. In K. Lindberg and D.E. Hawkins (eds) *Ecotourism: A Guide for Planners and Managers*. The Ecotourism Society, North Bennington, Vermont, pp. 7–11.
- Wheeller, B. (1991) Tourism's troubled times: responsible tourism is not the answer. *Tourism Management*, 12, 91–96.
- Wheeller, B. (1993) Sustaining the ego. *Journal of Sustainable Tourism*. 1, 121–129.
- Wight, P. (2001) Ecotourists: not a homogenous market segment. In D.B. Weaver (ed.) *The Encyclopaedia of Ecotourism*. CABI Publishing, Wallingford, pp. 37–62.
- Wilson, J.C. and Garrod, B. (2003) Introduction. In B. Garrod and J.C. Wilson (eds) *Marine Ecotourism: Issues and Experiences*. Channel View Publications, Clevedon, pp. 1–11.

Chapter 2

An Introduction to Tourism–Environment Relationships

Andrew Holden

Introduction

In attempting to understand the relationship of tourism with the natural environment it is necessary to reflect that the tourism system incorporates a range of stakeholders, including governments, the private sector, non-governmental organisations (NGOs), communities and tourists. The common denominator of these stakeholders is that they all involve human agency and behaviour. Consequently, when analysing the relationship between tourism and the natural environment it is necessary to consider that the relationship is in fact an expression of human interaction with nature. Subsequently, the values that we attach to nature, our cultural construction of what it is and our ethical relationship with it, will be influential for determining the impacts of tourism upon it. This chapter explores these issues, explaining how tourism has negative and positive consequences for the natural environment and, through a reciprocal relationship, how tourism itself can be threatened by human-induced changes in the environment. The chapter then progresses to an analysis of the ethical relationship between tourism and the natural environment.

Impacts of Tourism on the Natural Environment

To provide context for exploring the impacts of tourism on nature, it is useful to think of tourism as a system that links the environments of places that people come from, with the ones that they go to. Although the focus of this chapter is not upon the influence that the home environment has on demand for tourism, it is evident that many tourists seek what are perceived to be ‘unspoilt’ or ‘natural’ environments in contrast to the urban environments they often come from. The mental construction of the ‘natural’ environments we desire to travel to is an essential component of tourism. The home and destination environments are linked by the transport component of tourism, of which there is an increasing focus upon its environmental impacts, notably the contribution of aviation to global warming. Within this system, the environmental impacts of tourism may subsequently be viewed as having varying spatial dimensions, from the local to the global. The

impacts may also be differentiated into two broad categories of 'negative' and 'positive'; that is determined as being harmful or beneficial to nature.

The evaluation of the balance of this relationship, between the positive and the negative, is reflective of the values attributed to nature. If nature is categorized as possessing primarily an instrumental value for the purpose of human benefit, then there is a propensity for its over-use. This likelihood is enhanced by the characteristics of some of the natural resources that tourism relies upon, which are representative of 'common pool resources' (CPRs). According to Ostrom et al. (1999), the term CPR refers to resources with various types of property rights, but where exclusion of beneficiaries is impractical on the basis of cost, or is at least very costly, and the exploitation of the resource by one person reduces the benefit for another.

A subsequent major threat to the well-being of CPRs exists from the mentality of 'finders-keepers', that is a rush to harvest and secure the benefits of the resource before someone else does (Hardin 1968). Typical CPRs used for tourism may include the oceans, the atmosphere, beaches, coral reefs, wetlands, forests, wildlife and mountains. CPRs are vulnerable to over-use (what is taken from them) and also to pollution (what is put into them).

Conversely, if nature is considered to possess a range of values beyond the instrumental, then our interaction with it may be more benign. Certainly, if the conservation of nature through tourism can be demonstrated to have an economic value, then from a development perspective the argument for its continued existence is stronger. The next section of the chapter considers the negative and positive environmental impacts of tourism.

Negative Impacts: Environmental Pollution, Ecosystem Degradation and Reduced Environmental Aesthetic

As stated in the introduction, the spatial dimension of the environmental impacts of tourism range from the global to the local. At a global spatial scale, the contribution of air transport to pollution and global warming is an issue of debate and media attention. Although the amount of CO₂ emissions from aviation is relatively low compared to other sources: the International Air Transport Association (IATA 2007) suggest it is 2 per cent, whilst Gössling and Peeters (2007) estimate it to be between 3.4 per cent to 6.8 per cent; it is the rapid growth in air transport and its increasing significance as a source of CO₂ emissions that is of concern. Indeed, several climatologists state that aviation is the fastest growing causal factor of climate change (Garman 2006) and, per passenger kilometre, aviation produces more CO₂ than any other form of transport (Dubois and Ceron 2006). The potential impacts of greenhouse gas emissions from aircraft are made more significant by the effects of 'radiative forcing', i.e. as they are emitted at altitude, aircraft emissions have a greater relative impact than they would at surface level. The IPCC (2007) estimates the net effect of these emissions, taking into account radiative forcing, to be 2.7 times the carbon consumed in fuel.

Concern over air transport is a consequence of its exponential growth since the 1950s, with the annual rate of passenger growth averaging 5 per cent to 6 per cent per annum over almost a 50-year period. This demand has more recently been ‘fuelled’ by a combination of rapid growth in the numbers of budget flights and rising demand for flights in countries with rapidly developing economies. For example, in the case of China, during the period of May 2006 to May 2007, domestic flights increased by 18 per cent, whilst international flights to and from the country increased by 17 per cent (McCarthy 2007). The extent to which technology can provide a solution to increasing CO₂ emissions is uncertain, as the exponential increase in demand for flying means that total aviation emissions continue to rise despite improvements in emission-reducing technology (IPCC 2007, Adams 2007). In the absence of major-step technological progress, such as hydrogen or biofuel powered commercially viable aircraft, aviation emissions will likely be a major contributor to greenhouse gas emissions in the future. Even if alternative fuel sources were available, their medium- to long-term environmental impacts are uncertain, and the removal of land from agricultural use for biofuel production is already having adverse economic effects leading to rising food prices and shortages.

Whilst aviation, and other carbon-based modes of transport used for tourism and tourism infrastructure, generally makes a contribution to global warming, in a reciprocal relationship, global warming poses a threat to the future of tourism. The World Tourism Organisation (2003: 8) state: ‘In two environments that are vital for tourism activities and where tourism is an equally vital component in regional and local economies – coastal zones and mountain regions – climate change puts tourism at risk’. Thus, environmental change can threaten the very nature of the destinations that tourists seek to visit using air travel (Holden 2005).

Alongside its global effects, tourism impacts upon the natural environment at a local level. Tourism development can directly threaten and destroy ecological habitats, although the scale of transformation is difficult to estimate in the absence of comprehensive detailed research. There is, however, empirical evidence that tourism can impact upon the functioning of ecosystems, for example coral reefs, which are broadly comparable in terms of species diversity to tropical rain forests (Hill and Hill 2001). Coral is threatened from its use for the construction of tourism facilities, inadequate sewage disposal from proximate tourism concerns and the operations of local tour enterprises. For example, coral is mined for building materials in Sri Lanka, India, the Maldives, east Africa, Tonga and Samoa (Mieczkowski 1995). In the Red Sea, sediment run-off as a by-product of tourism construction, combined with nourishment to enhance tourist beaches and poor quality sewage and rubbish disposal systems for tourism, threaten the well-being of an estimated 70 per cent of the reefs off the coast of Egypt (Goudie and Viles 1997). In turn, these changes challenge the viability of the area’s lucrative diving industry, which is dependent upon the attractive biodiversity of the reefs.

Further problems associated with the use of coral reefs are linked to tourist behaviour, including tourists walking upon it, divers breaking it through accidental

contact or purposefully for souvenirs, tour operators dragging their boat anchors through it and local people breaking it to sell or make handicrafts with. These activities result in mechanical damage to reefs, altering coral cover and morphology, and impacting upon coral diversity and productivity (Palaganas 1991, Hawkins and Roberts 1993, Roupheal and Inglis 1997, Jameson et al. 2007). Although tourism may pose a notable threat to coral reefs in some areas, it is important to keep its impacts in perspective. At a global level, other factors, including global warming, tropical storms, increased sedimentation from deforestation, industrial pollution and over-fishing, also pose significant threats to reefs (Holden 2005).

Another example of the pressure that tourism can place on ecosystems includes the deforestation of mountainsides for ski development. Mountain areas harbour ecosystems that are sensitive to change, with forest removal not only potentially resulting in a loss of habitat for wildlife, but also leading to increased overland flow after rain and reduced soil cohesion. The combination of increased overland flow and reduced soil cohesion can heighten the propensity for landslides and in some cases lead to subsequent loss of life. The deaths, for example, of 60 people in northern Italy and southern Switzerland in 1987 were caused by landslides, which were attributed to deforestation for ski development (Simons 1988).

Coastal wetlands, an ecosystem that covers 6 per cent of the earth's surface and is significant for its biodiversity, carbon storing qualities and flood control capabilities, may also be placed under pressure from tourism development (Holden 2008). The restricted availability of land for tourism development in coastal and island areas, combined with the engineering capability to reclaim wetlands, means that they are increasingly being used for tourism development. For example, the Sir Donald Sangster International Airport in Jamaica and the French Mediterranean resort of 'La Grande Motte' were built on reclaimed wetlands (Holden 2008). The loss of wetland ecosystems not only leads to a loss of flora and fauna, but also decreases the ability of land areas to cope with flooding and reduces the capacity of the earth's 'carbon sink'.

Alongside the problem of aircraft emissions already alluded to, other types of pollution related to tourism include water, noise and a reduced environmental aesthetic. Water pollution is a major problem in many tourist regions of the world. For instance, in the most visited tourist area of the world, the Mediterranean, only 30 per cent of over 700 towns and cities on the coastline treat sewage before discharging it into the sea (Jenner and Smith 1992). In the Caribbean Basin, where tens of millions of tourists annually join the 170 million inhabitants, only 10 per cent of the sewage is treated before being discharged into the sea. Although these figures for sewage treatment may appear low, compared to other areas of the world they are actually good. Other regular international tourist destinations such as east Asia, east Africa and the islands of the South Pacific, with a few exceptions, have either no sewage treatment or treatment plants that are totally inadequate for the size of the population (Jenner and Smith 1992).

Although nearly two decades after the writing of Jenner and Smith (1992) it could be expected that these figures should have improved, it is suggested that

the extent of any improvement is likely to be irregular, dependent upon political will and economic resources to mitigate the effects of pollution. The problem of water contamination from human sewage is not caused exclusively by tourism, but is reflective of an inadequate infrastructure to meet the needs of both local populations and tourists.

Alongside the consequences it can have for human health, causing diseases ranging from mild stomach upsets to typhoid through the intake of water contaminated by faeces, human sewage also causes eutrophication (nutrient enrichment) of water. The consequence of a serious health scare for tourism demand was dramatically demonstrated in 1988 in Spain, when the fear of a typhoid outbreak resulting from contaminated water led to a 70 per cent decline in tourist reservations the next year (Kirkby 1996). The process of eutrophication may also impact on tourism demand, as was the case on the Romagna coast of Italy, where the total number of tourist bookings received by accommodation establishments in the area fell by 25 per cent in 1989 compared to 1988, owing to the eutrophication of the Adriatic and the spread of algae on the surface of the water (Becheri 1991).

In addition to its contribution to air pollution, a further negative externality of air traffic is noise pollution (Mieczkowski 1995). Noise pollution is a particular problem for those residents who are juxtaposed to busy international and domestic airports, and the proposed development of airports may sometimes lead to opposition by local people and environmental protest groups, for example Narita Airport in Tokyo in the 1990s (Shaw 1993) and, more recently, Heathrow in London. Noise pollution from tourism may be particularly noticeable in destinations where tourists are searching for quietness and peace. Air flights in remote areas where a lack of noise is expected, such as the Grand Canyon in the US and the Himalayas, can cause disruption to tourists and recreationists. Noise pollution from the construction of tourism facilities can also be a problem for residents and tourists. Briguglio and Briguglio (1996) observe that intense noise is generated by the building of hotels and other construction activity in destinations. Moreover, night clubs open until the early morning and increased car traffic from tourism movements add to the noise pollution experienced by both residents and tourists in tourism destinations.

Although possibly more subjective than necessarily scientific, a further category of pollution is 'aesthetic pollution'. Commenting on the development of tourism in the Guadeloupe and Martinique islands in the Lesser Antilles, Burac (1996: 71) observes: 'The most worrying problem now prevalent in the islands relates to the anarchic urbanisation of the coasts ... Also, the built-up areas by the seaside are often not aesthetically attractive due to the diversity of architectural styles, the disappearance of traditional creole homes and the disorderly way in which public posters are displayed'. Similarly, four decades ago Mishan (1969: 141) wrote: 'Once serene and lovely towns such as Andorra and Biarritz are smothered with new hotels and the dust and roar of motorised traffic'.

Whilst it would be unrealistic to assume that tourism may develop without involving the transformation of nature, it is misleading to portray tourism as having a purely negative effect upon the natural environment. Similar to most other types of economic activity, tourism involves some degree of trade-off between the realization of human benefits and environmental costs. The demand for tourism can play a vital role in giving an economic value to a natural resource to remain in a conserved state. Tourism may also play a significant role in raising awareness to support the establishment of protected areas to conserve nature. This chapter now turns its attention to exemplifying such issues.

Positive Impacts: The Case for Tourism as an Agent of Conservation

Whilst nature has been a key component of recreational tourism since the epoch of Romanticism, the emergence of recent terminology, including ecotourism, nature tourism, sustainable and responsible tourism underlines the central role of nature to tourism. Although controversy exists over what these terms actually mean and how they should be practised, they are nevertheless significant to the context of thinking of tourism as an agent for conservation.

In contrast to other kinds of economic land-use, such as agriculture, forestry and mining, which typically require the modification or destruction of natural resources to realize their utility, user-benefits may be achieved through tourism from experiencing nature in a largely conserved state. The fact that tourists are willing to pay to travel to, and to use accommodation and other services to view what is held to be 'nature', brings varying degrees of economic benefits at a national and local level to a destination. Consequently, an economic value may be given to natural resources to remain in a conserved state, the link to sustainable tourism development being axiomatic, as suggested more generally by Pearce (1993: 15):

If, on the other hand, conservation and the sustainable use of resources can be shown to be of economic value, then the dialogue of developer and conservationist may be viewed differently, not as one of necessary opposites, but of potential complements.

Whilst comparative estimates of the economic value of natural resources in a conserved state that can be realized through tourism are limited, research in the Amboseli National Park in Kenya in the late 1980s found the economic value of a lion to be US\$27,000 per annum, measured against the tourists' willingness to pay to see it in its natural habitat (Boo 1990). In terms of relative economic use, the park's net earnings from tourism were found to be 50 times higher than the most optimistic projection for agricultural use.

The potential economic value of ecotourism vis-à-vis other development options was also emphasized in the case of the Mabira Forest Reserve in Uganda. Recognized for its biodiversity, the forest was under threat of removal for sugar cane plantation to produce ethanol. However, an economic analysis of the value

of ecotourism and carbon capture calculated an annual value of US\$316 million compared to US\$20 million for ethanol production (Smith 2007).

Although the revenues that can be obtained through tourism can help to protect the environment from other more environmentally destructive development alternatives, an important caveat of this argument is that if natural habitats or wildlife are judged not to have sufficient economic value in comparison to other development options, then a pretext is established for their removal.

Shackley (1996) highlights a further danger of the economic valuation process for the conservation of wildlife. There are many species of wildlife which are not attractive to tourists in the dramatic sense that elephants and lions may be, but they have an essential role to play in the functioning of the ecological system of the area. Invertebrates, for example, perform essential ecosystem functions, yet they are largely overlooked in terms of ecotourism planning (Huntly et al. 2005). Because such species are not on the tourists' itinerary of animals to view, their value will be undermined in economic terms, placing their continued survival under threat.

Where tourism takes place in spatially defined boundaries, such as national parks or other types of protected areas that possess entry points and an evident management structure, the application of economic methodology to assess the value of wildlife that can be realized through tourism will be easier to implement than in areas which are less well defined. The use that revenues from tourism can have in protected areas is varied, including the conservation of natural resources, funding for infrastructure, employment opportunities, supplementary livelihood opportunities and human development opportunities. In the Annapurna Conservation Area of Nepal, for example, tourism revenues have enhanced economic development, whilst also being used to aid conservation programmes (Holden and Ewen 2004).

The vulnerability of tourism demand to external forces, however, means that basing protected area policy upon revenues from tourism is an unstable and potentially risky option. External factors that influence the demand for tourism to protected areas include security concerns, competition from other national parks and destination areas, and changing market fashions (Font et al. 2005).

Alongside methodology aimed at calculating the value of the environment, policy instruments can also be adopted to help give recognition to the value of conserving environments. One way of aiding the conservation of natural resources in developing countries is through 'debt-for-nature swaps'. The basis of this approach is that a conservation-based, non-governmental organisation (NGO) buys off an amount of a country's national debt, in return for guarantees that the indebted country will support and manage the conservation of a designated area, such as a national park. This will usually involve the development of appropriate management plans for the park. In effect, the NGO is purchasing the 'development rights' or, alternatively, the 'no-use' rights to a particular area of a country. As Pearce (1993) points out, debt-for-nature swaps are the only way in which estimates of the 'existence' (that is non-use) value of nature have been established.

According to Stone (1993), the concept of the debt-for-nature swap is an extension of the proposal made by Jomo Kenyatta, the founder of modern Kenya, at a conference in 1961. He intimated at the time that if African wildlife was a world possession, then 'the world could pay for it'. The process is based upon the buying of national debts on the world's money market at a fraction of their face value; for example, an NGO may pay a bank 25 cents for every dollar of a government debt. Banks are willing to discount the value of the loan and debt because of the risk that an indebted country may default upon its payment completely. As a consequence, debt-for-nature swaps usually involve the co-operation of national government, international and local NGOs and banks. In essence, a government is giving up part-sovereignty of a designated area of its country to an NGO.

Environmental Ethics

Debt-for-nature swaps are reflective of a philosophy that recognizes values in nature that transcend purely the instrumental and economic. A theme of this chapter is that as tourism is a phenomenon of human agency, how the tourism–environment relationship manifests itself will be reflective of how stakeholders value nature. The values that are held about nature extend beyond the economic, to incorporate the recreational and aesthetic; values that are especially significant in the case of tourism. Holmes Rolston III (1988) recognizes a range of other values that are held by nature, including the scientific, genetic, historical, cultural–symbolic and character-building. At a time when changes in the surrounding environment are forcing us to reconsider our position relative to nature, tourism as a sub-system of society will also adjust correspondingly to reflect new environmental considerations, priorities and ethics.

The relevance of environmental ethics to the relationship between tourism and nature can be illustrated through the following question: 'Should safaris be based upon the "shooting" of wildlife, as they predominantly were at the beginning of the twentieth century, or the "observing" of wildlife, as they are now?' In attempting to answer this question it is necessary to consider our relative position to nature and our consequential 'duty' towards it.

Reflections on the human position relative to nature were expressed by the influential western philosopher René Descartes in the seventeenth century. Arguing that unlike humans, animals possessed no souls and minds, he believed animals to be irrational and insensible making them incapable of suffering. This separation of the human and non-human worlds is dualistic, creating a pretext for the use of nature to satisfy human desires and aspirations. Descartes encouraged the belief that humans are the masters of nature. The dualism inherent to Cartesian philosophy was also advocated by another seminal thinker of western thought, Immanuel Kant (Ponting 1991). The basis of Kantian philosophy was to treat nature as non-divine, giving humanity a spiritual freedom and domination over it.

The application of Cartesian philosophy to practice implies that nature lacks an intrinsic value, possessing only an instrumental one, thus logically humans have no obligation of duty towards how nature is used. A dramatic example of this philosophy applied to the context of tourism is General Franco's policy for tourism development in Spain in 1959. Entitled the *Plan Nacional de Estabilization*, it possessed an inherent creed of 'crecimiento al cualquier precio' or 'growth at any price'. Similarly, the development of Cancun in Mexico from a village housing 12 Maya families in the 1970s to a resort receiving over 2.5 million visitors per annum, resulting in the destruction of rainforests and mangroves, the filling of lagoons and the levelling of sand dunes (Lynas 2003), is representative of the application of this ethic in tourism.

Whilst Descartes and Kant can be seen to have been influential to attitudes towards nature, not all philosophers subscribed to such a dualistic narrative. Expressing a view of a spiritual connectedness between humans and other species, Benedict Spinoza believed in 'animism' or 'organicism', that a single and continuous force permeated all beings and things, a manifestation of God-created substance. Likewise, Henry Moore, an important philosopher of the seventeenth century, believed that the spirit of God or *anima mundi* was present in every part of nature (Holden 2008).

In more contemporary times, although concerns for the effects of industrial development were expressed by the Romantic movement in nineteenth century Europe, and the effects of capitalism by Henry Thoreau and John Muir during the same century in America, it was Aldo Leopold who overtly expressed the human–nature relationship as an ethical issue in his seminal *A Sand County Almanac* (1949). In his call for a 'land ethic', he observes: 'In short, a land ethic changes the role of *Homo sapiens* from conqueror of the land-community to plain member and citizen of it. It implies respect for his fellow-members, and also respect for the community as such' (1949: 219). Leopold was suspicious of recreation and tourism, viewing it as having a negative effect on wildlife, whilst he perceived the travel trade as encouraging access to nature in bulk and reducing opportunities for solitude.

Concerns over the natural environment and the moral standing of nature have led to a large array of views on environmental ethics that cannot be categorized easily (Booth 1998). Based upon the 'rights' of nature to an existence, Simmons (1993) deduced two broad meanings of the application of ethics to the environment. These are an ethic for the 'use of the environment', which may be summarized as adopting an anthropocentric viewpoint of how we decide to use the resources of the earth, and an ethic 'of the environment', in which all non-human beings are given the same moral standing as that of the human species.

The first of these interpretations is probably the one most people, including decision-makers, would relate to and base their judgements upon. For example, the emphasis of sustainable development and sustainable tourism is very much upon environmental conservation for the well-being of future generations of humans (Cater 2006, Saarinen 2006), rather than concerns of the intrinsic right of

nature to an existence. The implication of Simmons' (1993) second position is that if other sentient and non-sentient beings have the same moral standing as humans, they should also have equal rights to those of the human world. Applied to law, this would mean, for example, that a hotel owner could be sued on behalf of the species occupying a coral reef if sewage emissions from the hotel were harming it. If the coral reef species won, the hotel would have to find another means of sewage disposal, as the rights to existence would be paramount.

There is evidence that an ethical viewpoint is influencing the action of tourism stakeholders on the basis that some existing patterns of behaviour can harm the environment. Notably, codes of conduct have been developed by various organisations in the private and public sectors to help guide appropriate tourist behaviour (Holden 2005). Garrod and Fennell (2004), for example, comment that there has been the development of a plethora of codes of conduct for whale-watching over recent years. The private sector seems to be placing emphasis upon demonstrating corporate social responsibility and assessing the performance of companies according to the 'triple bottom line' (Elkington 1998), considering their social, environmental and economic impacts.

The extent that an environmental ethic has or has not permeated into the consumer market for tourism is debatable. Given the growth of green and responsible consumerism it would seem logical to expect evidence of this trend within the tourism market. Certainly, the advent of prefixes such as eco-, sustainable and responsible tourism over the last twenty years would seem to be indicative of such a trend. However, the extent of the willingness of consumers to take adaptive measures to mitigate the negative environmental impacts of tourism, for example by paying higher prices to support environmental conservation schemes or flying less, will only become evident with time and the monitoring of environmental trends within the market.

Conclusion

The relationship that exists between tourism and the environment is a complex one to evaluate because of the range of ethics, meanings and spatial dimensions that are inherent to it. The emphasis of much of what has been written about this relationship, both in academia and the media, tends to assume that any damage to nature as a consequence of tourism activity should be regarded as harmful. Conversely, any use of tourism to ensure nature conservation should be regarded as positive and beneficial. Yet, this dichotomy of the interaction of tourism with nature is in many cases a false one, as the balance of this interaction is not a simple case of good or bad. For example, for tourism to act as an agent of conservation and to alleviate poverty in developing countries, usually necessitates the movement of tourists from developed countries by means of air travel, with the consequent contribution of aircraft emissions to the process of global warming. Similarly, the establishment of protected areas for wildlife conservation, which tourists may

choose to visit, may entail the forceful separation of indigenous peoples from their land and natural resources essential to their livelihoods, as with the case of the Maasai in Kenya (Holden 2008).

In a global system that encompasses different cultural and ethical perspectives, it is unsurprising that there will be a range of different viewpoints on the benefits or costs of the interaction between tourism and nature. These differences incorporate the construction of the meaning of ‘environment’, at its most dualistic ranging from an external reality that can be scientifically theorized or measured, to an entity possessing spiritual connections with other sentient and non-sentient beings, connected to our past and future.

Our cultural constructions of nature will subsequently influence the range of values and sense of duty we attach to nature, determining our perceptions of the levels of acceptable change in the environment for the purpose of tourism development. For example, in emphasizing a spiritual connection with the environment, the first concern of the First Haudenosaunee chiefs in North America when making decisions about the use of nature is to respect the needs of the ‘seventh generation’ (Holden 2008). This refers to the seventh generation of the past, who sacrificed their own needs to ensure that natural resources were available for the present generation, alongside the seventh generation of the future. Consequently, the First Haudenosaunee may hold different views about tourism development than the non-Haudenosaunee person. Similarly, the traditions of Mayan culture believe that every human has an animal counterpart and vice-versa, thus to harm one is to harm the other.

Our views on the meaning and significance of the environment are not likely to remain static, however, changing with time. Our existing preconceptions may be challenged by several factors, including: scientific discovery, such as global warming; philosophical reflection and debate, such as other sentient beings possessing emotions and consequently having ‘rights’; or through new spiritual awakenings, such as the re-discovery of pagan belief systems of nature. Such paradigm shifts are likely to result in a re-evaluation of our ethical relationship with nature. At present, there exists no universal agreement about the environmental ethic of tourism’s interaction with nature. Holidays, for example, are advertised that encourage tourists to participate in seal-culling, the killing of baby seals by club or bullet, in Newfoundland and Norway (Jowit and Soldal 2004). From an ethical perspective, it may be argued that whilst the individual right of the seals to exist is being violated, the killing of seals is justified in an eco-holistic framework to maintain fish stocks and the viability of the ecosystem. Of course, how the seals are killed, the enjoyment individuals may attain from it and the financial profits that are attributed from it, add additional complications to this argument.

However, the evaluation of the balance of the interaction of tourism with nature, as either positive or negative, will influence tourism policy and the dynamics of the market. Presently, the dominant paradigm of tourism development is sustainability, supported by a conservation ethic, derived from a realisation that our future well-being and survival is dependent upon the surrounding natural environment. This

is a significant shift from the instrumental use of nature that characterised tourism development for most of the twentieth century. The next environmental paradigm shift may be centred upon a spiritual connection to our surroundings, leading to recognition of the rights of nature, and having legal and economic implications for tourism development. Subsequently, the relationship of tourism with nature may be viewed as one of transition, reflecting the changing environmental values of society.

References

- Adams, D. (2007) Flights reach record levels despite warnings over climate change. *The Guardian*, 9 May, 3.
- Becheri, E. (1991) Rimini and Co – the End of a Legend? Dealing with the Algae Effect. *Tourism Management*, 12, 229-235.
- Boo, E. (1990) *Ecotourism: The Potentials and Pitfalls* (Vol. 1). World Wide Fund for Nature, Washington.
- Booth, D.E. (1998) *The Environmental Consequences of Growth*. Routledge, London.
- Briguglio, L. and Briguglio, M. (1996) Sustainable Tourism in the Maltese Isles. In L. Briguglio, R. Butler, D. Harrison and W.L. Filho (eds) *Sustainable Tourism in Islands and Small States: Case Studies*. Pinter, London, pp. 161–179.
- Burac, M. (1996) Tourism and Environment in Guadeloupe and Martinique. In L. Briguglio, R. Butler, D. Harrison and W.L. Filho (eds) *Sustainable Tourism in Islands and Small States: Case Studies*. Pinter, London, pp. 63–74.
- Cater, E. (2006) Ecotourism as a Western Construct. *Journal of Ecotourism*, 5, 23–39.
- Dubois, G. and Ceron, J.P. (2006) Tourism/Leisure Greenhouse Gas Emission forecasts for 2050: Factors for change in France. *Journal of Sustainable Tourism*, 14, 171–191.
- Elkington, J. (1998) *Cannibals with Forks: The Triple Bottom Line of 21st Century Business*. New Society Publishers, London.
- Font, X., Cochrane, J. and Tapper, R. (2005) *Pay per Nature View: Understanding Tourism Revenues for Effective Management Plans*. World Wide Fund for Nature, Netherlands.
- Garman, J. (2006) If I were ... Aviation Minister. *The Ecologist*, December, 23–24.
- Garrod, B. and Fennell, D.A. (2004) An analysis of whalewatching codes of conduct. *Annals of Tourism Research*, 31, 334–352.
- Gössling, S. and Peeters, P. (2007) 'It does not harm the environment!' An analysis of industry discourses on tourism, air travel and the environment. *Journal of Sustainable Tourism*, 15, 402–417.
- Goudie, A. and Viles, H. (1997) *The Earth Transformed: An Introduction to Human Impacts on the Environment*. Blackwell, Oxford.

- Hardin, G. (1968) Tragedy of the Commons. *Science*, 162, 1234–1238.
- Hawkins, J.P. and Roberts, C.M. (1993) Effects of recreational scuba diving on coral reefs: trampling on reef-flat communities. *Journal of Applied Ecology*, 30, 25–30.
- Hill, J.L. and Hill, R.A. (2001) Why are tropical rain forests so species rich? Classifying, reviewing and evaluating theories. *Progress in Physical Geography*, 25, 326–354.
- Holden, A. (2005) *Tourism Studies and the Social Sciences*. Routledge, London.
- Holden, A. (2008) *Environment and Tourism* (2nd edition). Routledge, London.
- Holden, A. and Ewen, M. (2004) Issues of natural resources, social well-being and tourism in the Annapurna area of Nepal. *ASEAN Journal on Hospitality and Tourism*, 3, 51–63.
- Huntly, P.M., Van Noort, S. and Hamer, M. (2005) Giving increased value to invertebrates through ecotourism. *South African Journal of Wildlife Research*, 35, 53–62.
- IATA (2007) *It's growing. But it will still be small in 2050*. Available at: <http://www.iata.org/whatwedo/environment/campaign/bonsai> [accessed: 11 July 2007].
- IPCC (2007) *Climate Change 2007: The Physical Science* (Basic summary for policymakers). International Panel on Climate Change, Geneva.
- Jameson, S.C., Ammar, M.S.A., Saadalla, E., Mostafa, H.M. and Riegl, B. (2007) A quantitative ecological assessment of diving sites in the Egyptian Red Sea during a period of severe anchor damage: a baseline for restoration and sustainable tourism management. *Journal of Sustainable Tourism*, 15, 309–323.
- Jenner, P. and Smith, C. (1992) *The Tourism Industry and the Environment*. The Economist Intelligence Unit, London.
- Jowit, J. and Soldal, H. (2004) It's the new sport for tourists: killing baby seals. *The Observer*, 3 October, 3.
- Kirkby, S.J. (1996) Recreation and the quality of Spanish coastal waters. In M. Barke, J. Towner and M.T. Newton (eds) *Tourism in Spain: Critical Issues*. CABI Publishers, Wallingford, pp. 190–211.
- Leopold, A. (1949) *A Sand County Almanac*. Oxford University Press, Oxford.
- Lynas, M. (2003) Should I stay or should I go? *The Geographical*, December, 100–104.
- McCarthy, M. (2007) Green groups dismayed as flights soar to record high. *Observer*, 8 May, 11.
- Mieczkowski, Z. (1995) *Environmental Issues of Tourism and Recreation*. University Press of America, Lanham, MD.
- Mishan, E.J. (1969) *The Costs of Economic Growth*. Penguin, Harmondsworth.
- Ostrom, E., Burger, J., Field, C., Norgaard, R., and D. Policansky (1999) Revisiting the Commons: Local Lessons, Global Challenges. *Science*, 284, 278–282.

- Palaganas, V.P. (1991) Anchor damage on the coral reef of Sombrero Island, Batangas, Philippines. *Coastal Zone: Proceedings of the Symposium on Coastal and Ocean Management*, 4, 3318–3329.
- Pearce, D. (1993) *Economic Values and the Natural World*. Earthscan Publications, London.
- Ponting, C. (1991) *A Green History of the World*. Sinclair-Stevenson, London.
- Rolston III, H. (1988) *Environmental Ethics: Duties to and Values in the Natural World*. Temple Press, Philadelphia.
- Rouphael, A.B. and Inglis, G.J. (1997) Impacts of recreational SCUBA diving at sites with different reef topographies. *Biological Conservation*, 82, 329–336.
- Saarinen, J. (2006) Traditions of Sustainability in Tourism Studies. *Annals of Tourism Research*, 33, 1121–1140.
- Shackley, M. (1996) *Wildlife Tourism*. International Thomson Business Press, London.
- Shaw, S. (1993) *Transport: Strategy and Policy*. Blackwell, Oxford.
- Simmons, I.G. (1993) *Interpreting Nature: Cultural Constructions of the Environment*. Routledge, London.
- Simons, P. (1988) Apres ski le deluge. *New Scientist*, 1, 46–49.
- Smith, L. (2007) Eco-tourists save forest jewels from bulldozers. *The Times*, 29 October, 29.
- Stone, C.D. (1993) *The Gnat Is Older than Man: Global Environment and Human Agenda*. Princeton University Press, Princeton.
- World Tourism Organisation (WTO) (2003) *Climate Change and Tourism*. World Tourism Organisation, Madrid.

Chapter 3

Thinking Globally about Ecotourism

Impact: The Contribution of Ecological Footprint Analysis

Colin Hunter

Introduction

Using Ecological Footprint analysis of simplified ecotourism scenarios, this chapter addresses the need to look beyond the immediate destination area when considering the environmental impact and sustainability of ecotourism, highlighting international air travel as an area worthy of particular focus. The chapter draws heavily on Hunter and Shaw (2006, 2007), and the reader is referred to these sources for more detailed explanation of Ecological Footprint analysis procedures and the ecotourism scenarios adopted.

Ecotourism is now apparently one of the fastest growing segments of the global tourism industry (Watkin 2003, Nyaupayne and Thapa 2004), and the popularity of ecotourism holidays has fuelled research interest in ecotourism, generating a substantial body of academic and policy literature (Fennell 2002a). Claims that ecotourism is pioneering the sustainable development of the tourism industry (for example de Villiers 2003) have thus come under growing scrutiny. Although environmental conservation is widely accepted as a key goal of ecotourism activity (for example Ross and Wall 1999, Buckley 2003a), some recent critiques of ecotourism (Mowforth and Munt 2002, Simmons and Becken 2004) have built upon earlier concerns about its frequent reliance on long-haul air travel (Wall 1997).

The global climate change potential of aviation is certainly regarded as a major concern. Current and projected growth rates in air transport, and the uniquely large potential impact of greenhouse gases other than carbon dioxide emitted by aircraft engines into the upper atmosphere, recently led Chapman (2007), for example, to conclude that aviation is currently the most unsustainable mode of transport available. Recent work by Peeters et al. (2007) on tourism transport by EU citizens found that tourism provides the main motive for air transport, with long-haul flights causing a disproportionately high impact in terms of volume of greenhouse gas emissions. Indeed, Simmons and Becken (2004) estimated the carbon dioxide emissions of even a relatively short one-way international flight between Australia and New Zealand to be approximately equivalent to the total transport-related

carbon dioxide emissions of a hypothetical 20 day self-drive ecotour in New Zealand.

It may be, therefore, that even if there exists a strong environmental conservation ethic locally at many (even most) individual ecotourism destinations, and that this manifests itself in genuinely low impact ecotourism products, the environmental costs of getting to these destinations is such that the ecotourism sector exerts a significant impact on natural resource use and environmental change at the global scale. In order to test this hypothesis, an environmental impact indicator or metric is required that can express different types of impact generated at different stages in the life-cycle of a tourism product (specifically comparing transit-related with destination-related impact) in the same unit. The Ecological Footprint (EF) provides such an indicator.

Ecological Footprint Analysis and Sustainable Tourism

First described in the early to mid 1990s (Rees 1992, Wackernagel and Rees 1996), the EF is a measure of demands upon the biological productivity and assimilative capacity of the biosphere imposed by a given human population over a certain time period (usually a year). The unique attribute of EF analysis is the expression of demand for all impact components as an equivalent land/sea area (in global hectares, or gha), thereby (it is claimed) facilitating comprehension of environmental impact and providing a powerful educational tool (for example Chambers et al. 2000, Wackernagel and Yount 2000). The EF itself does not exist in real space, but rather can be viewed as the aggregation of myriad actual land and sea areas around the world appropriated by a given population relying on the global movement of raw materials and products. Detailed descriptions of the procedures involved in EF analysis are found elsewhere (for example Wackernagel et al. 1999, Chambers et al. 2000). Typically, calculations account for and combine the use of energy (direct and embodied), foodstuffs, raw materials and water, and also capture transport-related impacts, the production of wastes (including carbon dioxide from the burning of fossil fuels), and the loss of productive land associated with buildings, roads and other aspects of the built environment. A much used benchmark for comparison in EF studies is the so-called 'fair earthshare' value; that is the global average area of productive land/sea space available annually on a per capita basis. One recent estimate of this, which excludes land set aside for non-human species, is 1.8 gha per year (World Wildlife Fund et al. 2004). Other cited values are slightly higher (Chambers et al. 2000, 2004), with around 2 gha per year regarded as a reasonable estimate (Venetoulis et al. 2004). It is important to appreciate that most advocates of the EF technique stress that EF calculations provide conservative estimates of global environmental impact (excluding, for example, the effects of toxic substances), and the tradition in EF analysis is to consciously err on the side of caution when making estimates of resource use and

waste production for use in calculations (for example Wackernagel and Rees 1996, Nijkamp et al. 2004).

The potential contribution of EF analysis as an indicator of sustainable tourism has been recognized (Hunter 2002). To date, however, tourism application of EF analysis has been very limited and largely focused on mass tourism. Cole and Sinclair (2002) used EF analysis to assess the sustainability implications of the advent of mass tourism to Manali in the Western Himalayas of India in the 1980s. They found that the EF of Manali increased by over 450 per cent between 1971 and 1995, with the tourism centre becoming increasingly dependent on ecosystem services external to the immediate area. This finding occurred despite international air travel by tourists not apparently being included specifically in the calculation. Gössling et al. (2002) estimated a leisure tourism EF for the Seychelles and found the per-tourist EF to be some 1.9 gha per year, with an average holiday in the Seychelles corresponding to between 17 per cent and 37 per cent of the annual EF of a citizen of an industrialized country. Well over 90 per cent of the total EF was found to be due to air travel to and from the destination. Work by the World Wildlife Fund UK (2002) presented EF analyses of two typical, two-week UK package holiday products from London to the popular Mediterranean destinations of Majorca and Cyprus. For Majorca, the total EF per tourist was found to be 0.37 gha, while the corresponding value for Cyprus was 0.93 gha. Accounting for approximately 50 per cent of the total EF in both cases, air travel was by far the largest single component of the holiday EF (albeit a much smaller proportion than that reported by Gössling et al. (2002), given the relatively short flights involved to the Mediterranean area). More recently, Peeters and Schouten (2006) estimated the EF associated with inbound tourism to Amsterdam for 2001. Transport components were found to dominate the overall EF estimate, accounting for some 70 per cent of the total, with air transport responsible for 87 per cent of the transport EF sub-total. The authors concluded that large reductions in the overall EF of inbound tourism to Amsterdam could be achieved by switching marketing effort from long-haul markets to short-haul markets, given the greater potential for modal shift away from air travel associated with the latter.

These examples illustrate the potential benefit of adopting the EF as a key environmental indicator of sustainable tourism: the EF provides a means of identifying and understanding globally-expressed demands on the biosphere brought about by tourism activity. As the studies demonstrate, this attribute of EF analysis may be particularly pertinent in accounting for the impact of tourism-related air travel to and from the destination. However, it would appear that in these studies the calculated tourism EF was interpreted as a wholly additional 'burden' on the global biosphere. In reality, whilst on holiday (and as recognized by the World Wildlife Fund UK in their study), the tourist is clearly not producing at home the footprint that would normally be created over the same period. The key indicator for any tourism product should therefore be the net, rather than the gross, EF generated, not least because the net estimate is more in keeping with the tradition in EF analysis of consciously erring on the side of caution when estimating

the magnitude of a particular activity or group (for example Wackernagel and Rees 1996, Monfreda et al. 2004, Nijkamp et al. 2004). The distinction between gross and net EF estimates may appear rather obvious, and in some scenarios the EF that would have been generated at home will be small compared to the gross holiday EF, meaning that the difference between gross and net EF values is insignificant. Yet, this will not always be so. In the case of ecotourism holiday products it may be particularly important to distinguish between net and gross EF estimates in assessing their apparent sustainability.

Ecotourists generally originate from developed countries where the average per capita EF is high, but frequently holiday in less developed countries (Gössling 1999) with considerably lower average per capita EFs (World Wildlife Fund et al. 2004). Theoretically, therefore, the potential exists for some international ecotourism holiday products to reduce, in net terms, the per capita EF of an individual whilst on holiday compared with the normal EF that would have been generated at home. The need to consider the EF of an international ecotourist in the context of the EF generated by her or him at home has been recognized by Fennell (2002b). Of course, the potential for a net EF reduction assumes that the tourist generates an EF whilst on holiday similar to the average per capita EF of the host population. Clearly, ecotourism activities may be relatively luxurious and resource intensive at the destination (Page and Dowling 2002). However, and in contrast to other types of tourism, products designed to be low impact at the destination occur frequently in the ecotourism sector (for example World Tourism Organization 2003). Thus, with ecotourism holidays in particular, there is at least the potential that net EF estimates will be substantially lower than gross EF estimates, as the rate of demand on natural resources at the destination compares favourably with that at home (as indicated by a large difference between home and destination EF values). Potentially, this difference may be large enough to outweigh the EF generated in transit (even by flying) between home and destination, giving an overall saving on natural resource demand associated with the ecotourism holiday.

Methodological Approach

EF analyses using primary data collected on tourist consumption of natural resources represents a very daunting challenge (Hunter 2002), and hundreds of EF analyses may be required in order to establish a range of EFs for ecotourism products. Using conservative and simplified hypothetical ecotourism scenarios, however, it is possible to establish an indicative range of estimates for the potential minimum net EF of ecotourism, thereby allowing tentative conclusions to be drawn about the likely global impact of ecotourism on natural resources. The findings reported below are not based on real ecotourism products. Rather, using conservative assumptions, the aim was to better understand the potential impact of ecotourism in global terms using hypothetical scenarios of relevance to the ecotourism sector. The ecotourism scenarios adopted paired leading source

Table 3.1 Average per capita Ecological Footprints of source and host countries (2001), and international tourist arrivals for host countries

Country	Ecological Footprint (gha)	Tourist Arrivals (2004)
<i>Source countries</i>		
France	5.8	-
Germany	4.8	-
Japan	4.3	-
United Kingdom	5.4	-
United States	9.5	-
<i>Host countries</i>		
Botswana	1.3	1,268,600
Brazil	2.2	3,797,200
Bulgaria	2.7	3,807,000
Costa Rica	2.1	1,237,000
Jordan	1.9	1,796,000
Kenya	0.9	874,800
Mexico	2.5	20,237,400
Mongolia	0.6	198,400
Morocco	0.9	4,929,000
Nepal	1.9	207,600
Peru	0.9	927,400
Phillipines	1.2	1,686,000
Senegal	1.2	457,400
Thailand	1.6	12,432,600

Source: Hunter and Shaw (2006)

countries involved in generating international tourists with geographically widespread host countries, ensuring variety in net EF estimates. Potential net EF estimates were made using readily accessible secondary data sources and an easily adopted calculation procedure as key attributes of a suitable indicator (Chambers et al. 2000, Bell and Morse 2003).

Five source countries were chosen, these being the leading spending nations on international tourism (USA, Germany, UK, Japan and France) according to the World Tourism Organization (2004). For the USA, two departure cities/airports

Table 3.2 One-way flight distances (km) between source and host countries

	France (Paris)	Germany (Frankfurt-am- Main)	Japan (Tokyo)	United Kingdom (London)	United States (New York)	United States (Los Angeles)
Botswana (Gaborone)	9,021	8,987	18,605	9,362	14,896	18,114
Brazil (Manaus)	11,243	11,630	21,256	10,978	5,656	7,632
Bulgaria (Sofia)	1,750	1,369	10,754	2,039	7,573	10,792
Costa Rica (San Jose)	9,171	9,958	13,179	8,906	3,556	4,878
Jordan (Amman)	3,358	3,022	9,116	3,661	9,195	12,414
Kenya (Nairobi)	6,397	6,462	15,981	6,838	12,372	15,590
Mexico (Mexico City)	9,195	9,453	11,245	8,897	3,365	1,349
Mongolia (Ulaanbataar)	9,345	9,793	3,058	9,310	14,844	11,677
Morocco (Marrakech)	2,126	2,574	11,827	2,293	7,827	11,046
Nepal (Kathmandu)	7,373	7,432	6,719	7,541	13,075	16,293
Peru (Lima)	11,600	11,987	15,755	11,335	5,876	6,736
Philippines (Manila)	10,738	10,290	3,050	10,944	14,190	12,801
Senegal (Dakar)	4,219	4,567	13,803	4,564	6,113	10,082
Thailand (Bangkok)	9,412	8,966	4,641	9,543	15,077	13,275

Source: Hunter and Shaw (2006)

Table 3.3 Procedure for calculating potential net Ecological Footprint estimates (gha per tourist per year)

Transit zone	
1.	Determine the total, round trip, flight distance (km).
2.	Obtain energy use per tourist (megaJoules, MJ) by multiplying flight distance by an energy intensity conversion factor of 2.0 MJ/pkm.
3.	Obtain the equivalent land area (ha of forest) per tourist (per year) required to sequester carbon dioxide production by dividing energy use per tourist by 73 GJ/ha (73,000 MJ/ha) (the number of gigaJoules that 1 ha of forest land will sequester, in carbon dioxide equivalent, per year when liquid fossil fuel is combusted) (World Wildlife Fund et al. 2000).
4.	Allow for the additional radiative forcing of aircraft emissions other than carbon dioxide emitted at altitude (Schumann 1994) by multiplying by a factor of 2.7 (International Panel on Climate Change 1999), giving a new estimate of required forest land (ha).
5.	Multiply by the appropriate 'equivalence factor' (in 2001 this was 1.38) to correct for forest land being more productive than average world space (Chambers et al. 2004), giving a final estimate of the transit zone per tourist footprint in gha per year.
Destination area	
6.	Use the host country average per capita EF as a proxy for the destination area EF of the tourist, reduced pro rata from an annualized value according to the length of stay.
Net EF	
7.	Use the average per capita EF of the source country and the length of stay away from home to calculate the per-tourist EF that would have been generated at home for the period away (again reduced pro rata from an annualized value), and subtract this from the gross per-tourist EF (the sum of steps 1 to 6).

were chosen (New York and Los Angeles) to reflect the size of the country and provide additional geographical diversity to EF estimates. Fourteen host countries were chosen: Botswana, Brazil, Bulgaria, Costa Rica, Jordan, Kenya, Mexico, Mongolia, Morocco, Nepal, Peru, Philippines, Senegal and Thailand. Unfortunately, no global initiative exists for the gathering of ecotourism data (The International Ecotourism Society 2000). Given the consequent lack of information on international ecotourist arrivals around the world, host countries were chosen on the following basis: they represent a geographically diverse group of destinations (at least one from each of the World Tourism Organization's world regions); and, all either provided the World Tourism Organization with information on ecotourism activities during the UN International Year of Ecotourism in 2002, are known to have established authorities to oversee ecotourism activities

(United Nations General Assembly 2003), or are well known as ecotourism destinations through case study material (for example Buckley 2003b). Selected host countries also offered some variety in terms of per capita EF characteristics. Per capita EF estimates for the source and host nations chosen are given in Table 3.1, along with international tourist arrivals for 2004, while flight distances are provided in Table 3.2. Distances were for direct flights where these were available, or assumed reasonably direct connections.

The gross tourism EF can be seen as having two broad components: that generated in the transit zone and that at the destination area. The net tourism EF is simply the sum of the transit and destination area components, less the source country EF for the period away from home. Potentially, there is the complication that a very small footprint will actually be generated by a tourist in the source country even when the tourist is away from home; for example, by leaving on some home heating or security lighting. The normal, major footprint components associated with energy use, transportation, food consumption and the consumption of other raw materials will be absent, however, so it would appear reasonable to assume the home-based footprint to be negligible during the international tourist trip.

The overall procedure for calculating a first order estimate of the potential annual equivalent net per capita (tourist) EF for international ecotourism involving air travel is summarized in Table 3.3. More detailed explanation of aspects of the procedure and the assumptions involved can be found in Hunter and Shaw (2007). The conservative nature of assumptions made implies that estimates should be regarded as potential minima.

Results

Using the above procedure it is possible to estimate the EF of hypothetical ecotourism scenarios using many hundreds of combinations of source and host countries. By way of illustration, the combination of the six source cities (in five countries) and fourteen destination countries chosen (Table 3.1), resulted in 84 net EF estimates being made for a particular length of stay period. Three lengths of stay were used (7 days, 14 days and 21 days) to provide added variety to the chosen source/host country combinations. The 21 day scenario may be particularly long for most ecotourism holidays (for example Page and Dowling 2002), but a long stay increases the likelihood of obtaining small potential net EF values, and it is important to allow for this theoretical possibility. (The reader is reminded that decreasing net EF estimates with increasing length of stay rests on the assumption that the EF generated by the tourist at the destination area is lower than that normally generated by her or him at home).

The US (Los Angeles/Mexico) scenario involved the shortest flight (some 1,349 km, one-way), with the longest one-way flight (at over 21,000 km) between Japan and Brazil (Table 3.2). Transit zone EF estimates (Table 3.4) ranged between

Table 3.4 Transit zone Ecological Footprint estimates (gha per tourist per year) between source and host countries

	France	Germany	Japan	United Kingdom	United States (NY)	United States (LA)
Botswana	1.84	1.83	3.80	1.91	3.04	3.70
Brazil	2.30	2.37	4.34	2.24	1.15	1.56
Bulgaria	0.36	0.29	2.20	0.42	1.55	2.20
Costa Rica	1.87	2.03	2.69	1.82	0.73	1.00
Jordan	0.69	0.62	1.86	0.75	1.88	2.53
Kenya	1.31	1.32	3.26	1.40	2.53	3.18
Mexico	1.88	1.93	2.30	1.82	0.69	0.28
Mongolia	1.91	2.00	0.62	1.90	3.03	2.38
Morocco	0.43	0.53	2.41	0.47	1.60	2.26
Nepal	1.51	1.52	1.37	1.54	2.67	3.33
Peru	2.37	2.45	3.22	2.31	1.20	1.38
Philippines	2.19	2.10	0.62	2.23	2.90	2.61
Senegal	0.86	0.93	2.82	0.93	1.25	2.06
Thailand	1.92	1.83	0.95	1.95	3.03	2.71

Source: Hunter and Shaw (2006)

0.28 gha (Los Angeles/Mexico) and 4.34 gha (Japan/Brazil). Minimum potential net EF estimates were obviously lower (Table 3.5). For a 7 day holiday these ranged between 0.15 gha (Los Angeles/Mexico) and 4.30 gha (Japan/Brazil); the latter value being equivalent to the average annual per capita EF of a Japanese citizen (Table 3.1). For a 14 day holiday, estimates ranged between 0.02 gha (Los Angeles/Mexico) and 4.26 gha (Japan/Brazil). In the former scenario, the holiday potentially accounted for only some 0.21 per cent of the average annual per capita EF of a US citizen. With the longer holiday period of 21 days, one scenario (Los Angeles/Mexico) actually produced a potentially negative net EF finding (-0.13 gha), suggesting the possibility of an overall reduction in demand on global renewable resources in this particular case. This was an isolated result, however, and over one-third (some 36 per cent) of the 14 day holiday scenarios, for example, produced potential net EF values greater than the (annual) 'fair earthshare' value of approximately 2 gha (Table 3.5). In only 20 per cent of 14 day scenarios did the holiday product potentially account for less than the annual average per capita footprint in low-income countries (0.8 gha according to World Wildlife Fund et al. 2004). Overall mean potential net EF estimates for the 7 day, 14 day and 21 day scenarios were 1.76 gha, 1.66 gha and 1.56 gha, respectively, giving a maximum difference of some 11 per cent. The size of the transit component (flight distance) was generally very important in determining the magnitude of potential net EF

Table 3.5 Potential net Ecological Footprint estimates (gha per tourist per year) for source/host country combinations

	France	Germany	Japan	United Kingdom	United States (NY)	United States (LA)
Botswana	1.75	1.76	<i>3.74</i>	1.83	<i>2.88</i>	<i>3.54</i>
	1.67	1.70	<i>3.69</i>	1.75	<i>2.73</i>	<i>3.39</i>
	1.58	1.62	<i>3.62</i>	1.67	<i>2.56</i>	<i>3.22</i>
Brazil	2.22	2.32	<i>4.30</i>	<i>2.18</i>	1.01	1.42
	<i>2.16</i>	<i>2.27</i>	<i>4.26</i>	<i>2.11</i>	0.87	1.28
	<i>2.10</i>	<i>2.22</i>	<i>4.22</i>	<i>2.06</i>	0.73	1.14
Bulgaria	0.30	0.24	<i>2.16</i>	0.37	1.41	<i>2.07</i>
	0.24	0.21	<i>2.14</i>	0.31	1.29	1.94
	0.19	0.17	<i>2.11</i>	0.27	1.16	1.81
Costa Rica	1.80	1.98	<i>2.65</i>	1.76	0.59	0.86
	1.73	1.93	<i>2.61</i>	1.69	0.45	0.72
	1.66	1.87	<i>2.56</i>	1.63	0.30	0.57
Jordan	0.62	0.57	1.82	0.69	1.74	<i>2.39</i>
	0.54	0.51	1.77	0.61	1.59	<i>2.24</i>
	0.47	0.45	1.72	0.55	1.44	<i>2.09</i>
Kenya	1.21	1.25	<i>3.20</i>	1.31	<i>2.36</i>	<i>3.02</i>
	1.12	1.17	<i>3.13</i>	1.22	<i>2.20</i>	<i>2.85</i>
	1.03	1.09	<i>3.06</i>	1.14	<i>2.03</i>	<i>2.68</i>
Mexico	1.82	1.89	2.26	1.76	0.56	0.15
	1.76	1.85	2.24	1.71	0.43	0.02
	1.69	1.79	<i>2.19</i>	1.65	0.28	-0.13
Mongolia	1.81	1.92	0.55	1.81	<i>2.86</i>	<i>2.21</i>
	1.76	1.89	0.53	1.76	<i>2.74</i>	<i>2.09</i>
	1.69	1.83	0.48	1.70	<i>2.59</i>	1.94
Morocco	0.34	0.46	2.35	0.39	1.44	<i>2.09</i>
	0.24	0.38	2.28	0.29	1.27	1.93
	0.15	0.30	<i>2.21</i>	0.21	1.10	1.76
Nepal	1.43	1.47	1.33	1.48	<i>2.53</i>	<i>3.18</i>
	1.31	1.36	1.23	1.35	<i>2.34</i>	<i>2.99</i>
	1.21	1.27	1.15	1.26	<i>2.15</i>	<i>2.81</i>
Peru	2.28	2.38	<i>3.15</i>	2.23	1.04	1.21
	<i>2.18</i>	<i>2.30</i>	<i>3.09</i>	<i>2.13</i>	0.87	1.05
	<i>2.09</i>	<i>2.22</i>	<i>3.02</i>	<i>2.05</i>	0.70	0.88
Philippines	<i>2.10</i>	<i>2.03</i>	0.56	<i>2.15</i>	<i>2.73</i>	<i>2.45</i>
	<i>2.02</i>	1.97	0.51	<i>2.07</i>	<i>2.59</i>	<i>2.30</i>
	1.93	1.89	0.44	1.99	<i>2.42</i>	<i>2.13</i>
Senegal	0.77	0.86	2.76	0.85	1.09	1.90
	0.69	0.80	2.71	0.77	0.94	1.75
	0.60	0.72	<i>2.64</i>	0.69	0.77	1.58
Thailand	1.84	1.77	0.90	1.88	<i>2.93</i>	<i>2.56</i>
	1.76	1.71	0.85	1.80	<i>2.73</i>	<i>2.41</i>
	1.68	1.64	0.79	1.73	<i>2.57</i>	<i>2.25</i>

Source: Hunter and Shaw (2006)

Notes: upper value = 7 day holiday, middle value = 14 day holiday, lower value = 21 day holiday; values greater than a 'fair earthshare' of 2 gha shown in *italics*.

estimates. This is clear by comparing the findings in Tables 3.2 and 3.4 with potential net EF estimates in Table 3.5. For example, and based on the particular scenarios adopted, the overall mean one-way flight distance to host countries was 11,356 km for Japan, 10,906 km for Los Angeles and 9,544 km for New York. These three source locations were associated with the largest number of scenarios where the potential net EF estimate exceeded a 'fair earthshare' value of 2 gha; 27 scenarios for Japan, 22 for Los Angeles and 18 for New York (Table 3.5). For the other source countries involved (France, Germany and the UK), overall mean one-way flight distance to host countries ranged between 7,496 km (France) and 7,606 km (Germany). For these countries, the number of scenarios where the 'fair earthshare' value was exceeded was much smaller; only seven or eight in each case (Table 3.5).

Discussion

Although these scenario-based estimates must be regarded as indicative at this stage, they do support anecdotal concern (for example Wall 1997) about the possible environmental impacts of flying, and flight distance, associated with increasing ecotourism activity. They also support the findings of previous relevant tourism EF applications in this regard, albeit based upon the use of net, rather than gross, EF estimates and using a variety of scenarios. Even for the 21 day scenarios (Table 3.5), the potential effect of differences between source and host country EF characteristics appeared small in reducing the influence of the transit EF component to overall net EF values. By considering net EF values and by using substantial lengths of stay for scenarios deemed to be relatively low impact at the destination, conservative estimates of relevance to the ecotourism sector have been calculated. Other assumptions in the estimation procedure also generally favoured the generation of conservative estimates, such that the values presented (Table 3.5) may be regarded as potential minima. Despite this, the findings suggest that ecotourism holidays involving air travel are likely to produce an absolute demand on global natural renewable resources. The magnitude of this demand may be very substantial: for a 14 day holiday involving return flight distances of 20,000 km or over, the potential net EF approaches, and then may greatly exceed, 2 gha (an approximation of the annual per capita 'fair earthshare' value).

Furthermore, findings assume that resource use by the ecotourist at the destination is relatively conservative, reflecting that of the host country. Yet, ecotourism activities may occur in much more up-market, resource-demanding contexts with tourists living much more luxurious lifestyles than locals (see, for example, Kontogeorgopoulos 2004). It could be argued, therefore, that in many circumstances, particularly involving popular or 'soft' forms of ecotourism (Page and Dowling 2002), it would be more appropriate to adopt the average per capita EF of the source country as a proxy for the EF generated at the destination (Gössling et al. 2002). Thus, the source country per capita EF would be used in both steps

6 and 7 of the calculation (Table 3.3), and the potential net tourism EF becomes the same as the transit zone EF provided by steps 1 through 5, irrespective of the length of stay. Consequently, the net EF can never be negative, and is simply a function of distance travelled by air. Changing the assumption about the nature of resource demand by ecotourists at the destination in this way produces higher potential net EF estimates, although increases are generally small. Comparing the 14 day findings, for example, in Table 3.5 with those in Table 3.4, mean values (1.66 gha and 1.86 gha, respectively) differ by only some 11 per cent (0.2 gha). Again, this reflects the general dominance of the transit component to overall potential net EF values in the scenarios. Thus, a higher EF at the destination area merely reinforces the key conclusion suggested by these findings: that ecotourism experiences involving international air travel will normally exert an absolute (and substantial) net demand on global natural resources.

Great caution is required, however, in drawing definitive conclusions on the sustainability of ecotourism from the estimates in Tables 3.4 and 3.5. The reader is reminded that findings are not based on real ecotourism products or the use of primary data on tourist natural resource demand, and EF analysis itself carries its own set of assumptions and simplifications (Nijkamp et al. 2004), adding to those made in the rudimentary estimation procedure for the net tourism EF given above. Also, our understanding of the radiative forcing effect of aircraft emissions at altitude remains incomplete (Sausen et al. 2005), and it is not clear how any additional radiative forcing effect should be incorporated into EF calculations. Furthermore, given that ecotourism emerged as an alternative to mass tourism, and much of the debate surrounding ecotourism has been driven by comparison with traditional mass tourism (for example Cater and Lowman 1994), one might base a judgement on the sustainability of ecotourism by comparing this sector with mass tourism. The key question now becomes, is ecotourism more or less demanding of natural resources, in EF terms, than mass tourism? At this stage, without detailed EF analysis for different product types, it is very difficult to provide a concrete answer, particularly as interpretations of ecotourism and ecotourism products may still differ widely (for example Simmons and Becken 2004). But because the ecotourism scenarios (supported by evidence from other studies) suggest a general dominance of the flight-related EF component to net EF estimates, it would only be logical to infer ecotourism as having a greater impact than mass tourism if, at a global scale, ecotourism products generate more air passenger kilometres than mass tourism products.

Unfortunately, there is a dearth of basic information on global ecotourism activity. According to the World Tourism Organisation, in the late 1990s all nature-related forms of tourism may have accounted for some 20 per cent of total international travel (The International Ecotourism Society 2000), but the contribution of ecotourism to this is unknown. It does seem clear, though, that long-haul air travel will increase: worldwide, it is predicted to grow faster, at 5.4 per cent per year over the period 1995 to 2020, than intraregional travel, at 3.8 per cent per year. Consequently, the ratio between intraregional and long-haul air travel

may shift from around 82:18 in 1995, to close to 76:24 by 2020 (World Tourism Organization 2004). Should it occur, this increase appears as likely to arise from mass market, package holiday products offered by the large, mainstream operators as from specialized ecotourism products provided by niche operators (for example Meyer 2003). Traditional beach holiday products to long-haul, exotic destinations, for example, are now offered routinely to UK residents (amongst the most frequent long-haul air travellers in the world) by mainstream operators and are increasing in popularity (Meyer 2003). The indirect evidence, therefore, suggests that now and in the near future ecotourism is very unlikely to generate more air passenger kilometres than conventional mass tourism.

Arguably, it is the apparently inexorable growth in international tourism involving air travel that is the fundamental problem, not the growth in one segment of this market. Although the findings presented in this paper suggest that the concerns expressed about ecotourism by some academic researchers may well be warranted on the grounds of transit zone environmental impact, the deeper lesson may be the need for more attention to be paid to transit zone impacts in general, irrespective of the associated product label. It should also be remembered, of course, that genuine ecotourism products carry a commitment at least to the protection of local ecosystems and other desirable actions. As yet, mass tourism products generally do not.

This said, there may be a worrying irony to be found in the assumed relationships between tourism activity, air travel and global climate change that is particularly striking in the case of the ecotourism sector. A focus on the non-consumptive use of local wildlife resources so embedded in principles of ecotourism and wildlife tourism may, even if fully adhered to, miss a larger point and fail to protect local ecosystems from the effects of climate change; effects in part derived from ecotourism-generated air travel itself. By way of illustration, cetacean (whale and dolphin) watching around Scotland's coastal waters has increased in importance in recent years and is ripe for further expansion (MacLellan 1999). Whales and dolphins are Scotland's leading wildlife attraction, providing up to 12 per cent of local income to some remote rural areas (Woods-Ballard et al. 2003). Although tempered with calls for strong local and national protection of cetacean species from the (localized) impacts of tourism activity, a lack of focused marketing effort and low foreign tourist numbers have been seen as important impediments to the more sustainable development of cetacean watching in Scotland. MacLellan (1999), for example, reports work which identified Germany and the USA as particularly noteworthy potential markets, while Woods-Ballard et al. (2003) even suggest manipulating the price structure of tourist trips to incentivize participation by those who travel a long distance. The irony, of course, is that evidence is already emerging of relatively rapid changes to the composition of local cetacean communities, around the north-west of Scotland at least, associated with climate change in the form of ocean warming, such that certain cetaceans may move out of areas of tourist significance, including areas specifically designated for marine conservation (MacLeod et al. 2005).

Conclusions

EF analysis permits a global perspective on tourism-generated natural resource demand using a consistent unit (gha) of evaluation, and it would appear to have much to offer in advancing our understanding of tourism environmental impact. Based on the simplified EF estimates presented above, it can be concluded that any tourism product or activity that relies on air travel, and particularly long-haul air travel, is likely to exert a substantial net demand on natural resource use and contribute to global climate change. This is likely to be the case even where, as with some ecotourism products, the rate of natural resource consumption by the tourist at the destination is markedly less than would be generated by her or him at home. Clearly, however, much remains to be accomplished in the application of EF analysis to tourism, and there is considerable scope for debate over the scenario-based assumptions and findings reported above in the context of ecotourism.

The use of proxy measures, and associated assumptions, is clearly an inherent weakness in the procedure reported in this chapter. Although difficult and time consuming, it is only through the collection and analysis of primary data for real ecotourism products that assumptions can be tested. These data need to include: the consumption of energy (including travel mode and distances), food/beverages, and water; other purchases (for example clothing, gifts); waste products of various kinds; and tourism-related buildings and other forms of infrastructure (Hunter 2002). Approaches to data collection might involve the keeping of diaries by tourists, questionnaire surveys of tourists, and information gathered from hotels, restaurants, bars, and the providers of excursions (Hunter 2002).

Of course, in order to accurately estimate the net EF of any given tourist/holiday, it would also be desirable to gather primary data to determine the normal, household EF of the tourist whilst at home, rather than relying on national average per capita data as a proxy. Household-level EF analysis is, however, in its infancy, although detailed household EF calculators do exist (Hunter et al. 2006), and these might also be adapted to transform resource consumption data from hotels and other tourist businesses into actual per-tourist EF estimates for the destination. As a partial alternative to intensive primary data gathering, one or both of the source and destination country per-tourist EF values could be estimated using more refined secondary data sources. A promising avenue for future research may be to utilize an input–output macroeconomic framework to redistribute and disaggregate national EF information down to the household and tourist economy levels using expenditure data (Wiedmann et al. 2005). Incorporating sensitivity analyses to this type of approach would allow a firmer basis for an understanding of (eco)tourism's natural resource demands, whilst avoiding the difficulties of gathering and utilizing primary data and selecting representative ecotourism products for analysis. Rather than relying wholly on primary data approaches, these might then be more focused on checking and/or calibrating estimates made using secondary data sources. Either way, however, a more concrete and detailed

understanding of the EF of (eco)tourism products does require the development of primary data approaches.

Although the ecotourism EF scenarios reported in this chapter suggest that most ecotourism products will exert a net demand on natural resource use and contribute to climate change globally, the ecotourism sector, given its current size, does not in all probability generate a total EF or demand on global natural resources as substantial as traditional forms of mass tourism. However, if ecotourism really is to act as a positive exemplar for the rest of the tourism industry, then there is a particular onus on those engaged in planning, delivering and promoting ecotourism products to consider the impact of these in more global terms and account for air travel natural resource demands in assessing their sustainability. More prosaically, natural ecosystems cannot be immune from the pressures exerted locally by global climate change, and therefore ecotourism products around the world must be regarded as at risk from climate change; change brought about in part by travel, particularly air travel, to the ecotourism destination.

References

- Bell, S. and Morse, S. (2003) *Measuring Sustainability: Learning from Doing*. Earthscan, London.
- Buckley, R. (2003a) Environmental inputs and outputs in ecotourism: geotourism with a triple bottom line? *Journal of Ecotourism*, 2, 76–82.
- Buckley, R. (2003b) *Case Studies in Ecotourism*. CABI Publishing, Wallingford.
- Cater, E. and Lowman, G. (1994) *Ecotourism: A Sustainable Option?* John Wiley & Sons, Chichester.
- Chambers, N., Simmons, C. and Wackernagel, M. (2000) *Sharing Nature's Interest: Ecological Footprints as an Indicator of Sustainability*. Earthscan, London.
- Chambers, N., Griffiths, P., Lewis, K. and Jenkin, N. (2004) *Scotland's Footprint: A Resource Flow and Ecological Footprint Analysis of Scotland*. Best Foot Forward, Oxford.
- Chapman, L. (2007) Transport and climate change: a review. *Journal of Transport Geography*, 15, 354–367.
- Cole, V. and Sinclair, J. (2002) Measuring the ecological footprint of a Himalayan Tourist Centre. *Mountain Research and Development*, 22, 132–141.
- de Villiers, D. (2003) Introduction. In World Tourism Organization (ed.) *Sustainable Development of Ecotourism: A Compilation of Good Practices in SMEs*. World Tourism Organization, Madrid, pp. 9–11.
- Fennell, D. (2002a) Ecotourism: where we've been; where we're going. *Journal of Ecotourism*, 1, 1–6.
- Fennell, D. (2002b) *Ecotourism Programme Planning*. CABI Publishing, Wallingford.
- Gössling, S. (1999) Ecotourism: A means to safeguard biodiversity and ecosystem functions? *Ecological Economics*, 29, 303–320.

- Gössling, S., Borgstrom Hansson, C., Horstmeier, O. and Saggel, S. (2002) Ecological footprint analysis as a tool to assess tourism sustainability. *Ecological Economics*, 43, 199–211.
- Hunter, C. (2002) Sustainable tourism and the touristic ecological footprint. *Environment, Development and Sustainability*, 4, 7–20.
- Hunter, C. and Shaw, J. (2006) Applying the ecological footprint to ecotourism scenarios. *Environmental Conservation*, 32, 294–304.
- Hunter, C. and Shaw, J. (2007) The ecological footprint as a key indicator of sustainable tourism. *Tourism Management*, 28, 46–57.
- Hunter, C., Carmichael, K. and Pangbourne, K. (2006) Household ecological footprinting using a new diary-based data gathering approach. *Local Environment*, 11, 307–327.
- International Panel on Climate Change (1999) *Aviation and the Global Atmosphere. A special report of IPCC working groups I and III*. Cambridge University Press, Cambridge.
- Kontogeorgopoulos, N. (2004) Ecotourism and mass tourism in southern Thailand: spatial interdependence, structural connections, and staged authenticity. *GeoJournal*, 61, 1–11.
- MacLellan, R. (1999) An examination of wildlife tourism as a sustainable form of tourism development in north west Scotland. *International Journal of Tourism Research*, 5, 375–387.
- MacLeod, C.D., Bannon, S., Pierce, G., Schweder, C., Learmonth, J., Herman, J. and Reid, R. (2005) Climate change and the cetacean community of north-west Scotland. *Biological Conservation*, 124, 477–483.
- Meyer, D. (2003) The UK outbound tour operating industry and implications for pro-poor tourism. *Pro-Poor Tourism Working Paper No. 17*. Overseas Development Institute, London.
- Monfreda, C., Wackernagel, M. and Deumling, D. (2004) Establishing national natural capital accounts based on detailed ecological and biological capacity accounts. *Land Use Policy*, 21, 231–246.
- Mowforth M. and Munt, M. (2003) *Tourism and Sustainability: Development and New Tourism in the Third World* (2nd edition). Routledge, London.
- Nijkamp, P., Rossi, E. and Vindigni, G. (2004) Ecological footprints in plural: a meta-analytic comparison of empirical results. *Regional Studies*, 38, 747–765.
- Nyaupane, G.P. and Thapa, B. (2004) Evaluation of ecotourism: a comparative assessment in the Annapurna conservation area project, Nepal. *Journal of Ecotourism*, 3, 20–45.
- Page, S. and Dowling, R. (2002) *Ecotourism*. Prentice Hall, Harlow.
- Peeters, P. and Schouten, F. (2006) Reducing the ecological footprint of inbound tourism and transport to Amsterdam. *Journal of Sustainable Tourism*, 14, 157–171.
- Peeters, P., Szimba, E. and Duijnisveld, M. (2007) Major environmental impacts of European tourist transport. *Journal of Transport Geography*, 15, 83–93.

- Rees, W.E. (1992) Ecological footprints and appropriated carrying capacity: what urban economics leaves out. *Environment and Urbanization*, 4, 121–130.
- Ross, S. and Wall, G. (1999) Ecotourism: towards congruence between theory and practice. *Tourism Management*, 20, 123–132.
- Sausen, R., Isaksen, I., Grewe, V., Hauglustaine D., Lee, D.S., Myhre, G., Köhler, M.O., Pitari, G., Schumann, U., Stordal, F. and Zerefos, C. (2005) **Aviation radiative forcing in 2000: an update on IPCC (1999)**. *Meteorologische Zeitschrift*, 14, 555–561.
- Schumann, U. (1994) On the effect of emissions from aircraft engines on the state of the atmosphere. *Annales Geophysicae*, 12, 365–384.
- Simmons, D.G. and Becken, S. (2004) The cost of getting there: impacts of travel to ecotourism destinations. In R. Buckley (ed.) *Environmental Impacts of Ecotourism*. CABI Publishing, Wallingford, pp. 15–23.
- The International Ecotourism Society (TIES) (2000) *Ecotourism Statistical Fact Sheet 2000*. Available at: <http://www.ecotourism.org/research/stats/files/stats.pdf>.
- United Nations General Assembly (2003) *Assessment of the results achieved in realizing aims and objectives of the International Year of Ecotourism (2003)*. Available at: <http://www.world-tourism.org/sustainable/IYE/IYE-Rep-UN-GA-2003.pdf>.
- Venetoulis, J., Chazan, D. and Gaudet, C. (2004) *Ecological Footprint of Nations*. Redefining Progress, Oakland, CA.
- Wackernagel, M., Onisto, L., Bello, P., Linareas, A.C., Falfan, I., Garcia, J., Guerrero, A. and Guerrero, C. (1999) National natural capital accounting with the ecological footprint concept. *Ecological Economics*, 29, 375–390.
- Wackernagel, M. and Rees, W.E. (1996) *Our Ecological Footprints: Reducing Human Impact on the Earth*. New Society Publishers, Gabriola Island, BC.
- Wackernagel, M. and Yount, J.D. (2000) Footprints for sustainability: the next steps. *Environment, Development and Sustainability*, 2, 21–42.
- Wall, G. (1997) Is ecotourism sustainable? *Environmental Management*, 21, 483–491.
- Watkin, J. (2003) The Evolution of Ecotourism in East Africa: From an Idea to an Industry. *IIED Wildlife and Development Series No. 15*. International Institute for Environment and Development, London.
- Wiedmann, T., Minx, J., Barrett, J. and Wackernagel, M. (2005) Allocating ecological footprints to final consumption categories with input–output analysis. *Ecological Economics*, 56, 28–48.
- Woods-Ballard, A.J., Parsons, E.C.M., Hughes, A., Velandar, K., Ladle, R. and Warburton, C.A. (2003) The sustainability of whale-watching in Scotland. *Journal of Sustainable Tourism*, 11, 40–55.
- World Tourism Organization (2003) *Sustainable Development of Ecotourism: A Compilation of Good Practice in SMEs*. World Tourism Organization, Madrid.

World Tourism Organization (2004) *Tourism Highlights: 2004*. World Tourism Organization, Madrid.

World Wildlife Fund, United Nations Environment Programme, World Conservation Monitoring Centre, Redefining Progress and The Centre for Sustainability Studies (2000) *Living Planet Report 2000*. World Wildlife Fund, Gland, Switzerland.

World Wildlife Fund, United Nations Environment Programme, World Conservation Monitoring Centre and Global Footprint Network (2004) *Living Planet Report 2004*. World Wildlife Fund, Gland, Switzerland.

World Wildlife Fund-UK (2002) *Holiday Footprinting: A Practical Tool for Responsible Tourism*. World Wildlife Fund-UK, Godalming.